

Anticancer Agents

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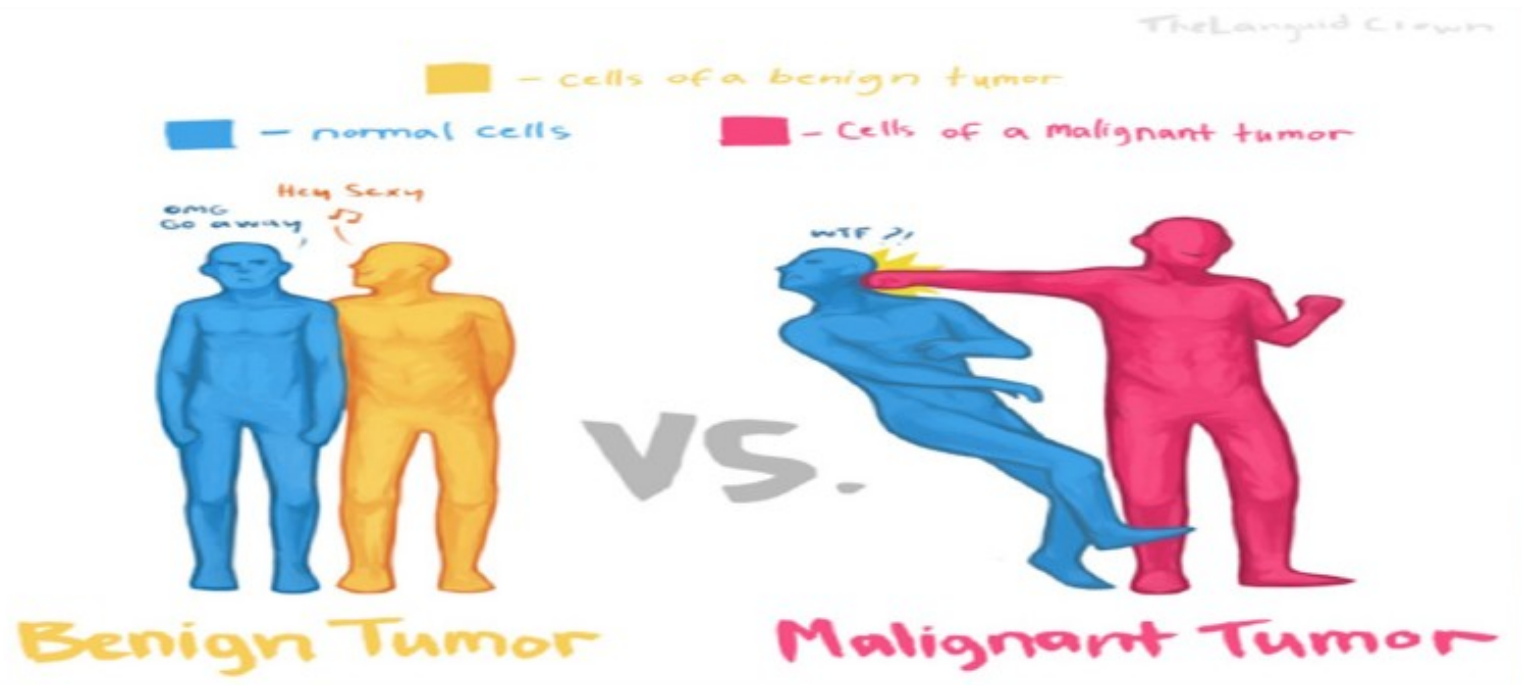
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Introduction

- Cancer is a disorder of cells in the body.
- It begins with a group of cells that fail to respond to the normal control mechanism and continue to divide without control.
- The new growth is called **tumor or neoplasia** and may be either *benign* or *malignant*.
 - A **benign tumor** is the one that remains localized whereas
 - A **malignant tumor** invades neighboring tissues, enters blood vessels, lymphatic vessels and other spaces
 - Can be carried to other areas of the body to form new tumors, “*metastasis*”.

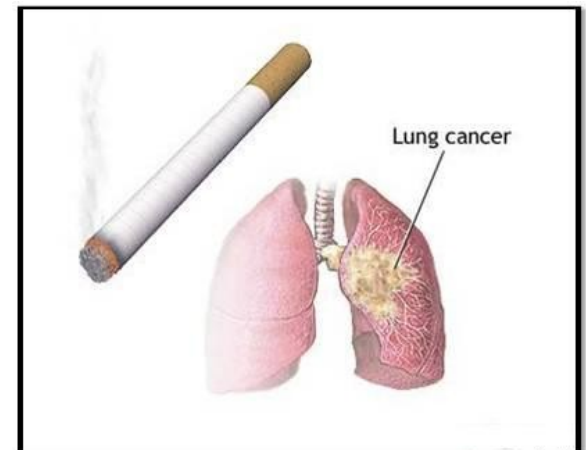
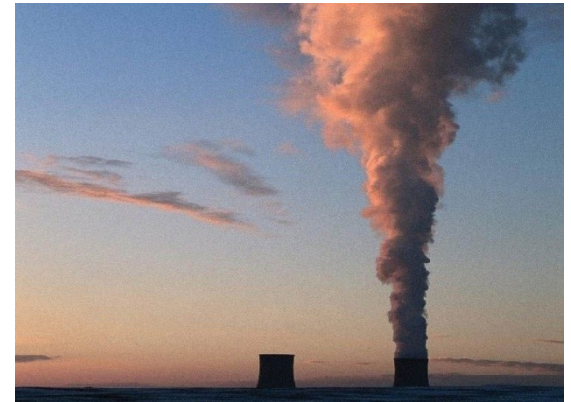
- **Benign:** non cancerous and not an immediate threat to life, even though treatment eventually may be required for health.
- **Malignant :** tending to worsen and cause death, invasive and metastasis



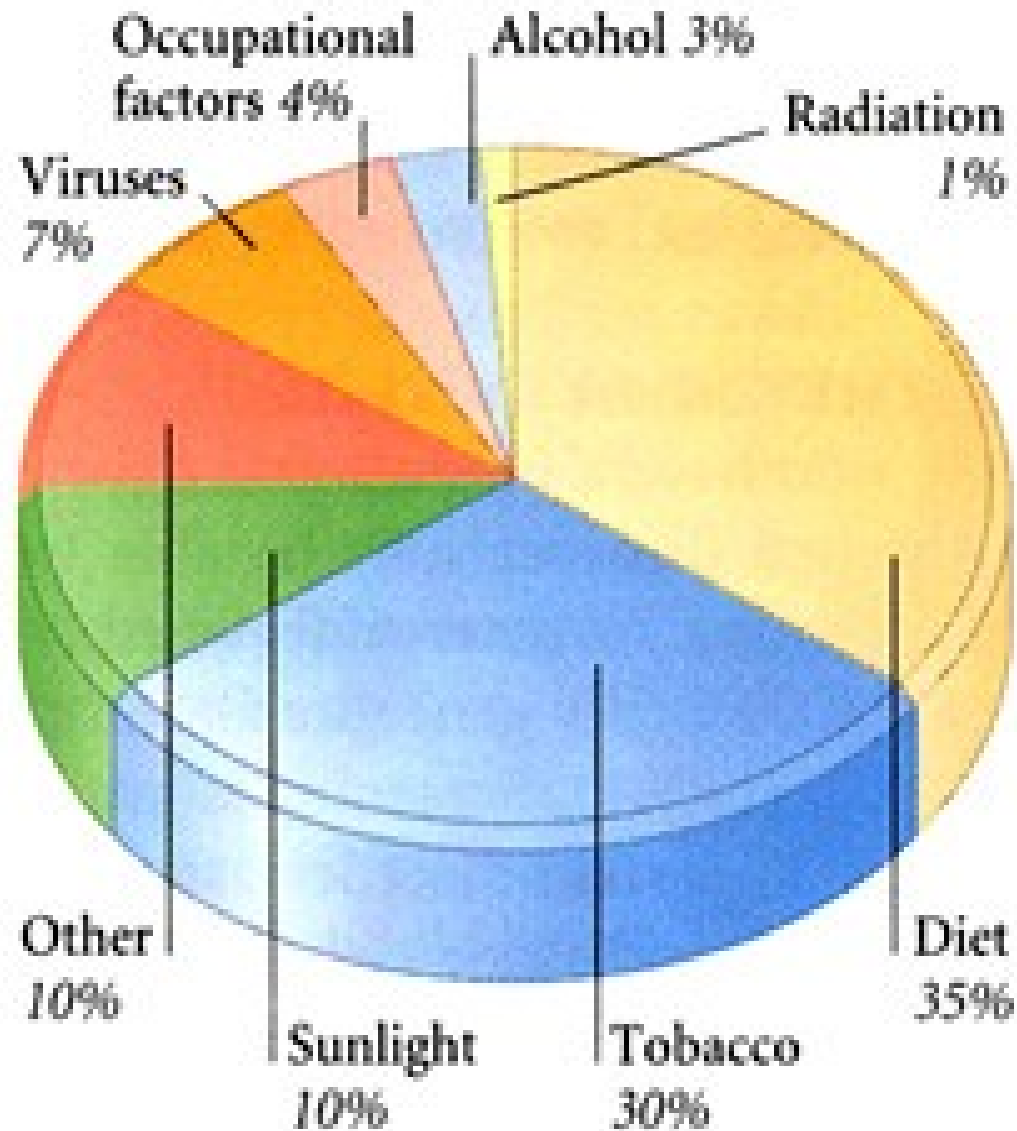
- Cancer is caused by abnormalities in cells.
- These abnormalities may be due to
 - Inherited genes from parents or
 - Outside exposures to the body such as chemicals, radiation or even infectious agents.
- The **defective genes** are called ***oncogenes***.
Substances which cause changes that can lead to cancer are called ***carcinogens***.

External causes of cancer

- Chemicals
 - Aromatic amines
 - Heavy metals
 - Smoke
 - Oxidizing agents (free radicals like ROS)
- Electromog (mobile)
- Radiation
- Infections
 - Viral infections



Causes of Cancer



Treatment strategies of cancer

- Surgery
 - Can be used for treatment and diagnosis of primary cancer
 - Recommended when we are sure to remove tumor with affecting vital organs
 - Limited to benign cancer
- Photoradiation
 - Using light of specific wavelength (620-640nm) to destroy tumor cells
 - Normal cells will not absorb light at this wavelength range

- Radiation
 - Using ionizing radiations (gamma-ray, beta-ray) to damage tumor cells
 - Radiation sensitizers
 - Razoxane
 - Dactinomycin
 - 5-fluorouracil
 - Limitations
 - Can also affect normal cells

- Gene therapy
 - To remove and replace defective genes
 - To replace defective tumor suppressor genes by genetically engineered genes
- Immunotherapy
 - To boost the immune system by using interferons
 - Interferons can increase the accuracy of immune cells to identify and destroy cancerous cells

- Chemotherapy
 - Using drugs to destroy cancer cells selcetively
 - Can be used along with other procedures to treat cancer
 - Can also localize tumor for surgery and radiation therapy

Chemotherapeutic Agents

- Classification
 - Alkylating agents
 - Antimetabolites
 - Hormones and Antihormones
 - Antibiotics
 - Plant products
 - Miscellaneous

Alkylating Agents

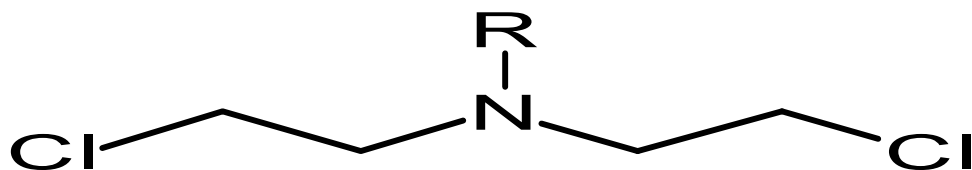
- Alkylating agents contain chemical groups which have the ability of forming covalent bonds with nucleophilic substances in the cell (DNA, RNA and certain enzymes).
- Most of the anticancer alkylating agents are bifunctional; possessing two alkylating groups.
- They are thus able to react with two groups and can cause **intra- or inter-chain cross-linking**, that can interfere with **transcription and replication of DNA**.

Alkylating Agents ...

- Classification of alkylating agents
 - Nitrogen mustards
 - Nitrosoureas
 - Aziridines
 - Sulfonic acid esters
 - Triazines
 - Platinum complexes

Nitrogen Mustards

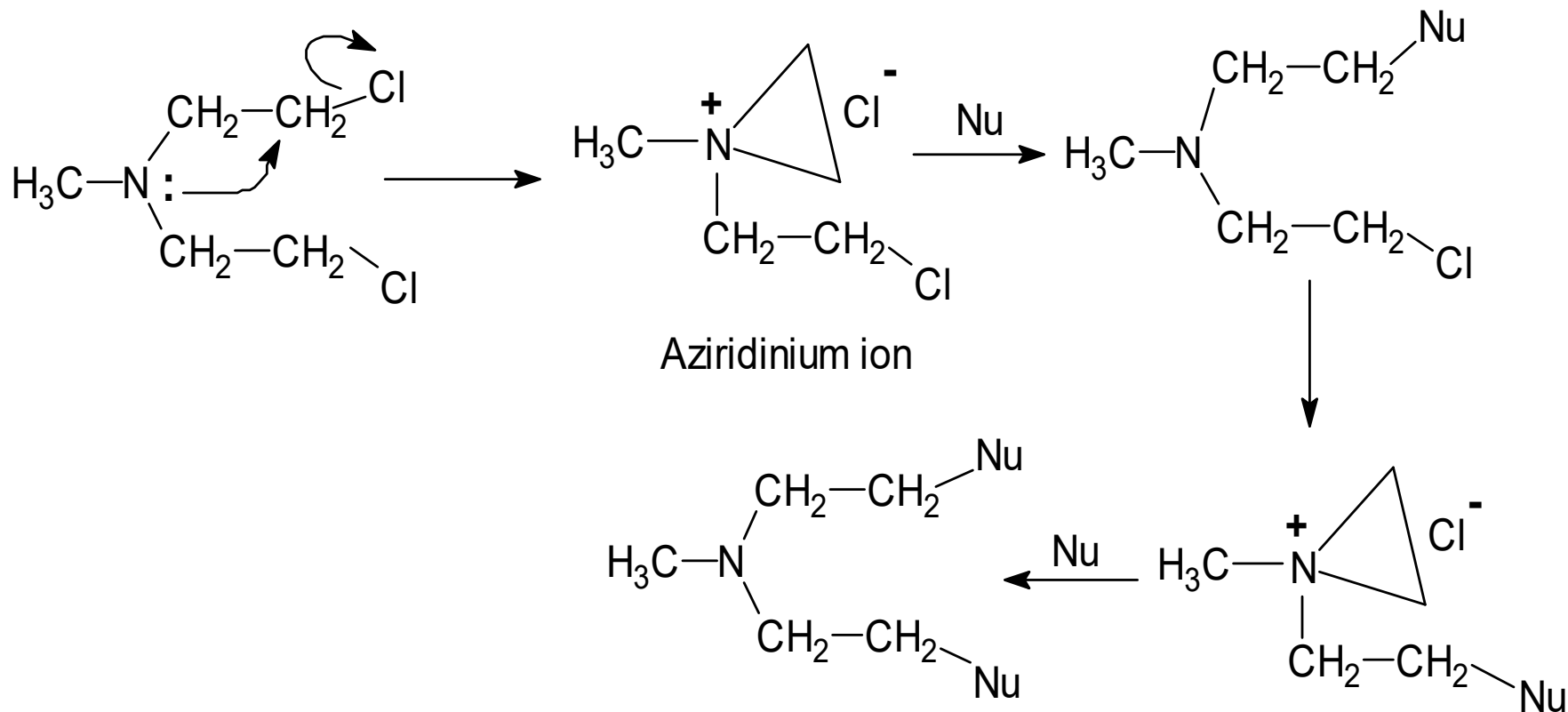
- Their original use stems from the observation during world war I, that individuals heavily exposed to mustard gas suffered damage to bone marrow and lymphoid tissues.
- **Mustard gas**: highly toxic, insoluble in water.
- **Nitrogen mustard** derivatives: less toxic, salts are solids and water soluble.



Mechanism of Action of Nitrogen Mustards

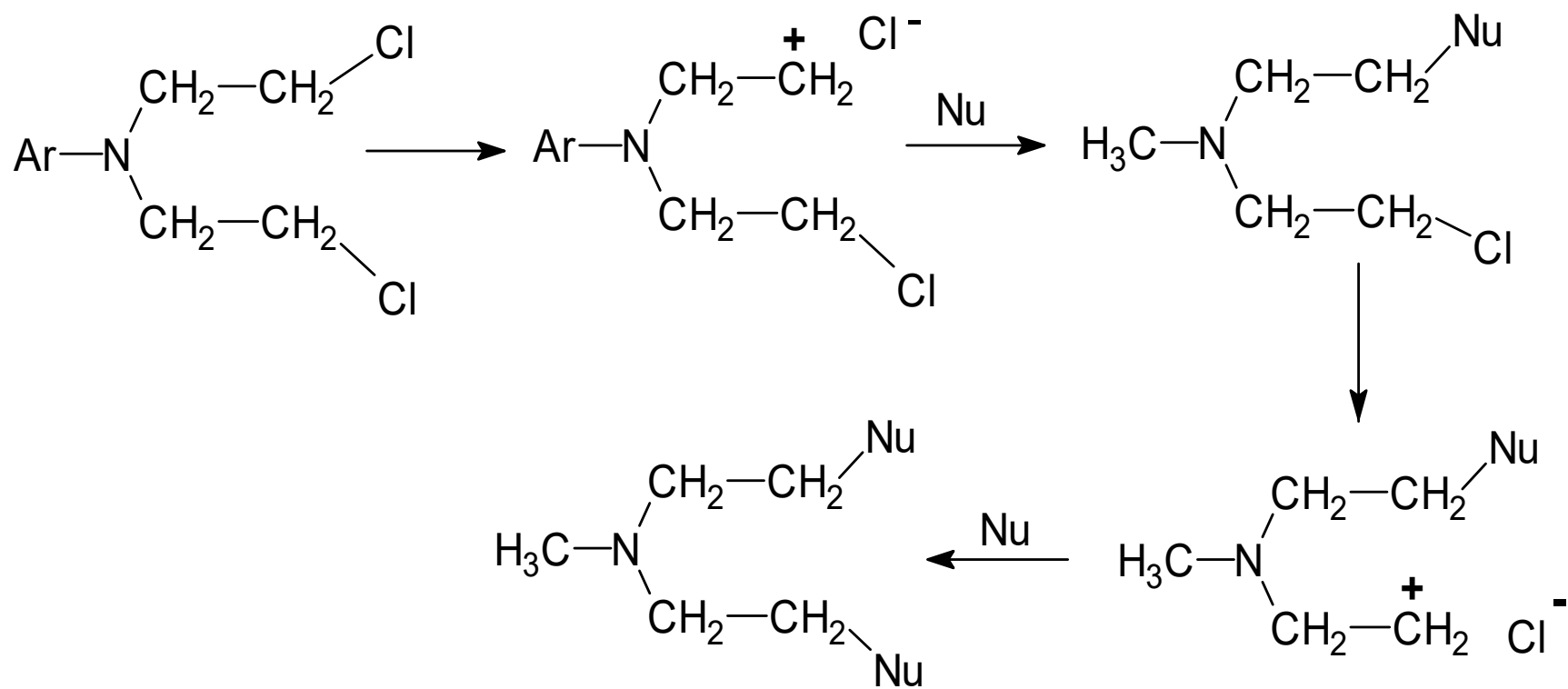
- **Aliphatic Substituted Nitrogen Mustards**
- The molecules undergo neighboring group reactions in which the nitrogen atom displaces a chlorine atom to give strained three-membered *aziridinium* intermediates.
- The highly strained three-membered ring rearranges by scission to the carbonium ion, which in turn reacts with the nucleophilic groups.
- The attack leads to *irreversible* covalent bond formation with the substrate.

MOA ...



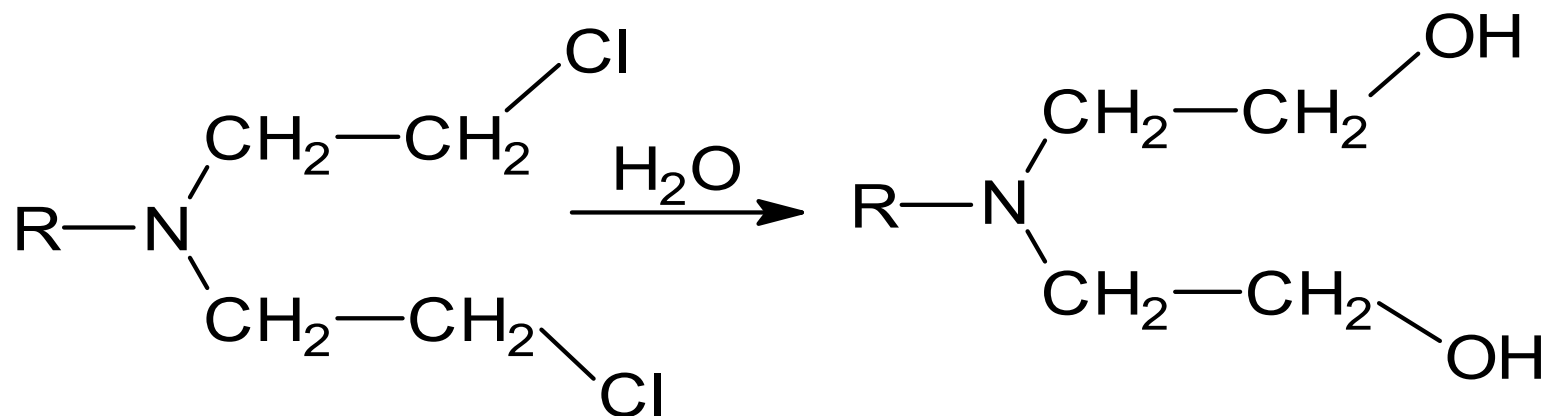
Nu = nucleophilic center rich in electrons (OH, NH₂, SH, COOH, SO₃H) present in enzymes, proteins RNA and DNA

- **Aromatic Substituted Nitrogen Mustards**
- Aryl substituted nitrogen mustards are relatively stable toward aziridinium ion formation, because the aromatic ring decreases the nucleophilicity of the N atom.
- The highly reactive carbonium ion species react with nucleophilic (electron-rich) sites on nucleic acids and proteins



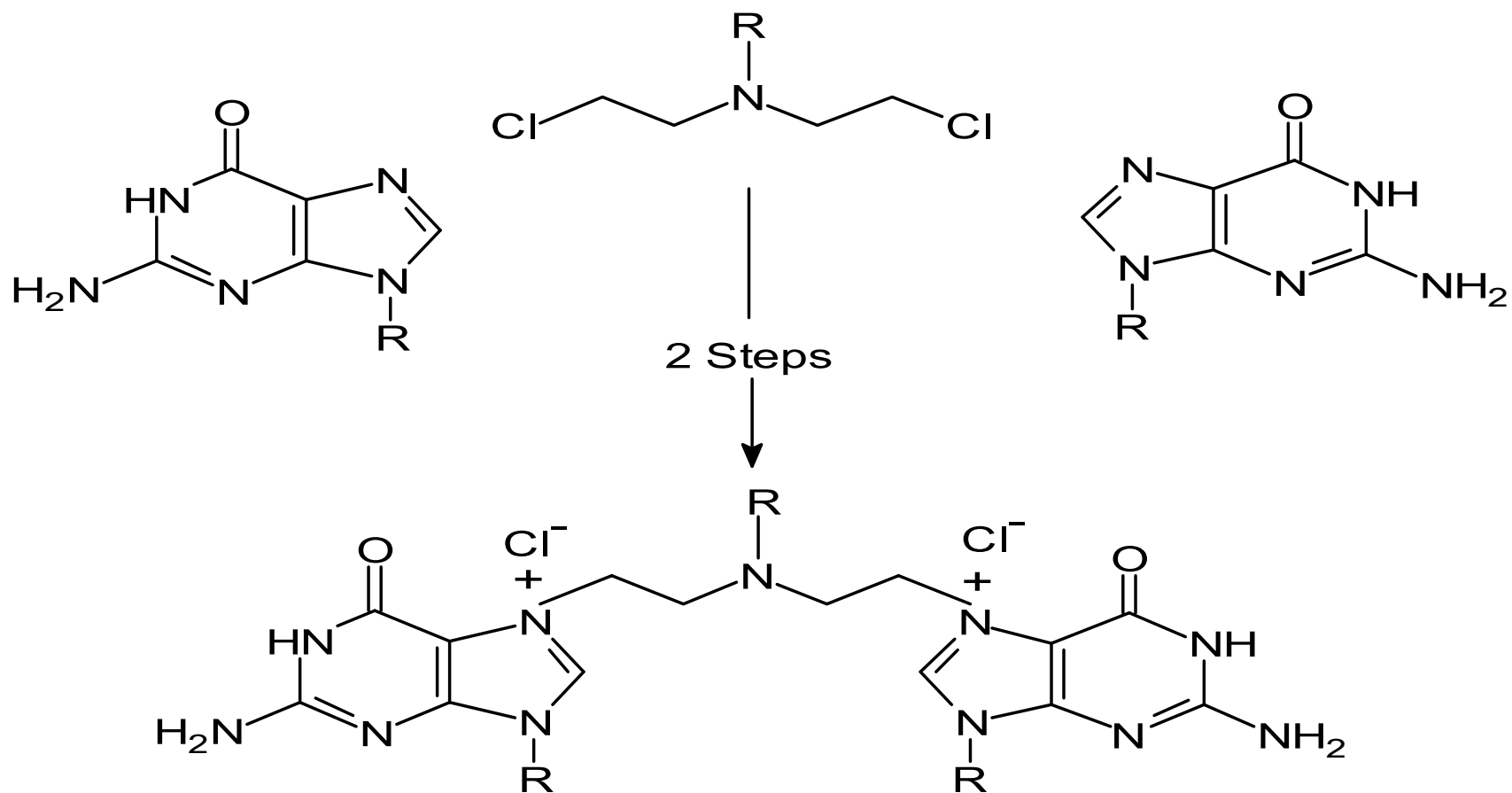
- **Stability in Water**

- An inactive product is formed. So, drugs are marketed as dried powders to be put in solution before use as I.V. infusion fluid.



Alkylation of DNA by Nitrogen Mustards

- In DNA, the N₇ of guanine, being strongly nucleophilic, is probably the main molecular target for alkylation in DNA.
- However N₁ and N₃ of adenine and N₃ of cytosine may also be affected.
- The alkylated structure has a positive charge in its imidazole ring which renders the guanine-ribose linkage susceptible to cleavage.
- Guanine is also implicated in the cross-linking of double helical DNA, which interferes with separation of the strands and prevents mitosis.



Structure-Activity Relationship

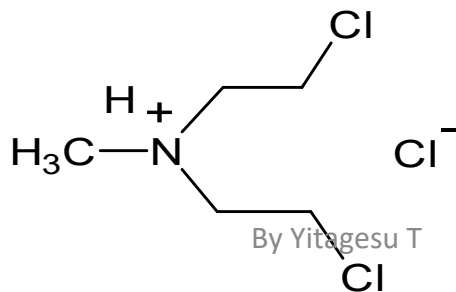
- Aliphatic nitrogen mustards are more soluble in lipid therefore more bioactive.
- The more localized the unshared pair of electrons on N the higher is the reactivity.
- Aromatic nitrogen mustards are of low basicity with a slow rate of C^+ formation therefore the molecules can reach distant sites.

SAR cont...

- The presence of an amino acid moiety with nitrogen mustard may direct the drug to a metabolic site affecting cancer.
- Mono-functional alkylating agents are less bioactive.
- Iodine or fluorine atoms instead of chlorine decrease the activity (slow formation of C^+ ion);
 - Bromine atoms lead to very reactive molecules but with high toxicity.

i. Mechlorethamine HCl (Mustine HCl)

- Used in the treatment of Hodgkin's diseases and other malignant neoplasm including lung carcinoma, chronic myelocytic leukemia and brain tumors.
- Adverse effects include bone-marrow depression, nausea and vomiting, neurological disorders, and liver disorders. It is administered intravenously and intracavitary.

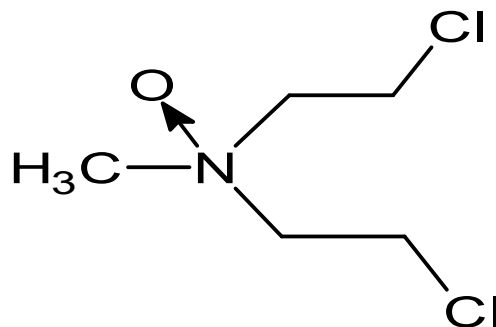


- **Hodgkin's disease** a disease with rubbery enlargement of lymph nodes and enlargement of spleen.



ii. Mustrone

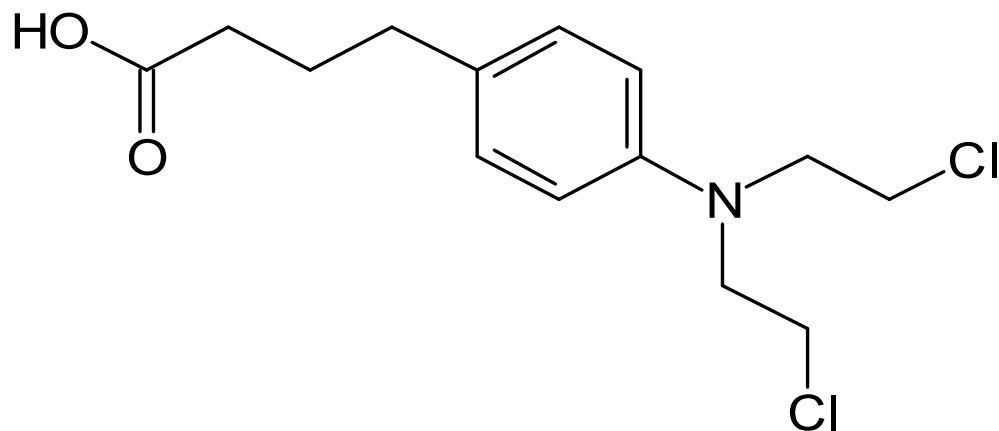
- The toxicity of the parent compound is greatly reduced with only slight reduction in activity.
- It is active after metabolic reduction to mechlorethamine.



iii. Chlorambucil

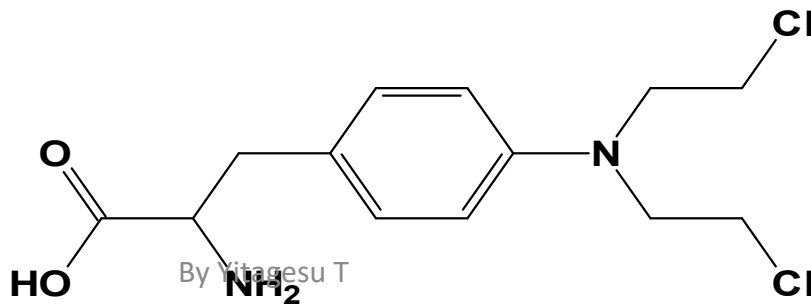
- An alkylating agent used in the treatment of leukemias and lymphomas and various other malignant neoplasms.
- It is the *slowest* acting and *least* toxic nitrogen mustard.
 - Aromatic nitrogen mustard
- Adverse effects include bone-marrow depression, GIT disturbances, neurotoxicity, lung disorders and sterility.
- It is administered orally.

Chlorambucil ...



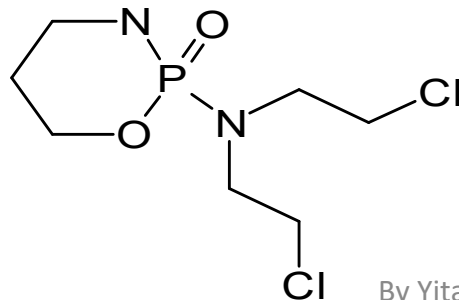
iv. Melphalan

- Phenylalanine-N-mustard
- Melphalan is used for its antineoplastic action in the treatment of **multiple myeloma** and in other malignant neoplasm including **tumors of the breast and ovary**.
- Adverse effects include bone marrow depression, rashes, and GIT disturbances. It is active orally.



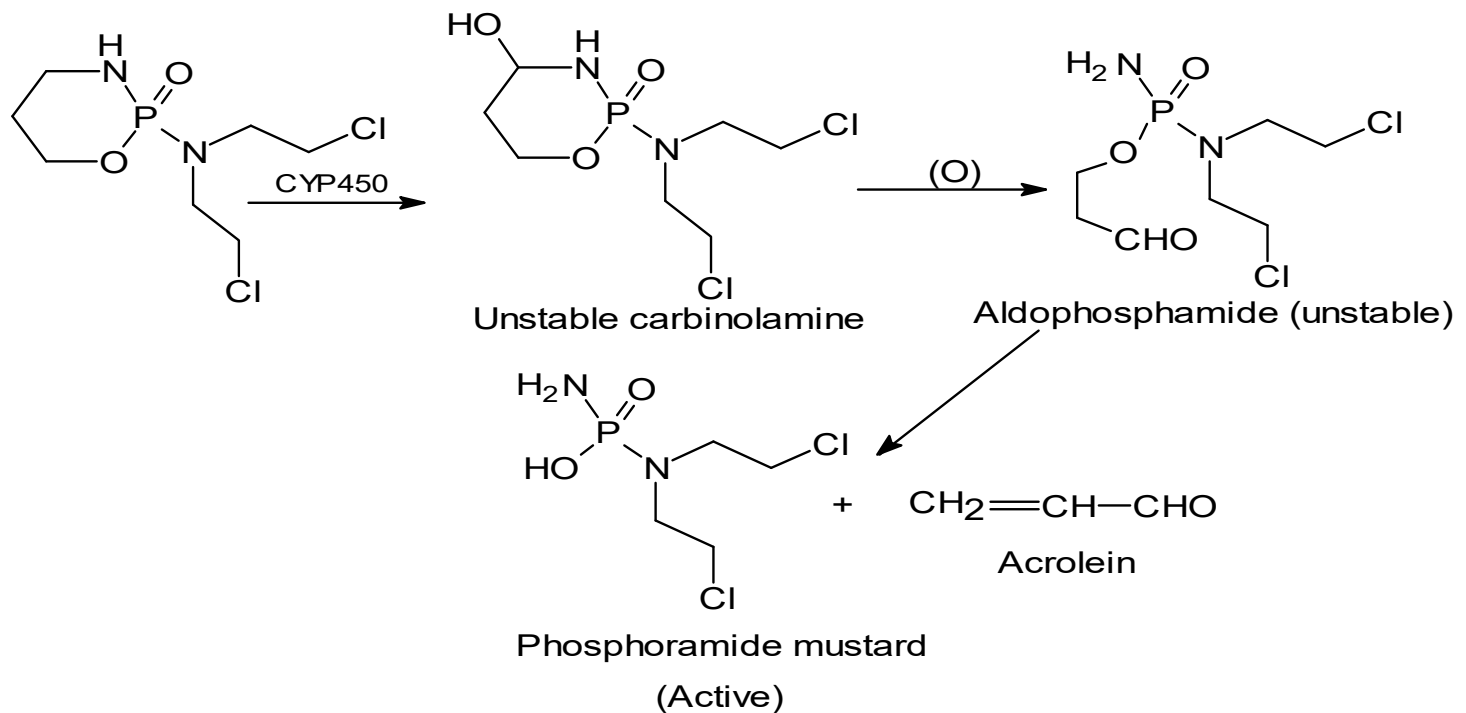
v. Cyclophosphamide

- Can be given by mouth or by intravenous injection:
- Although it is chemically related to the nitrogen mustards, the nucleophilicity of the mustard nitrogen is less likely to form an aziridinium ion than alkyl-substituted nitrogen mustards.
- Prodrug



Bio-activation of Cyclophosphamide

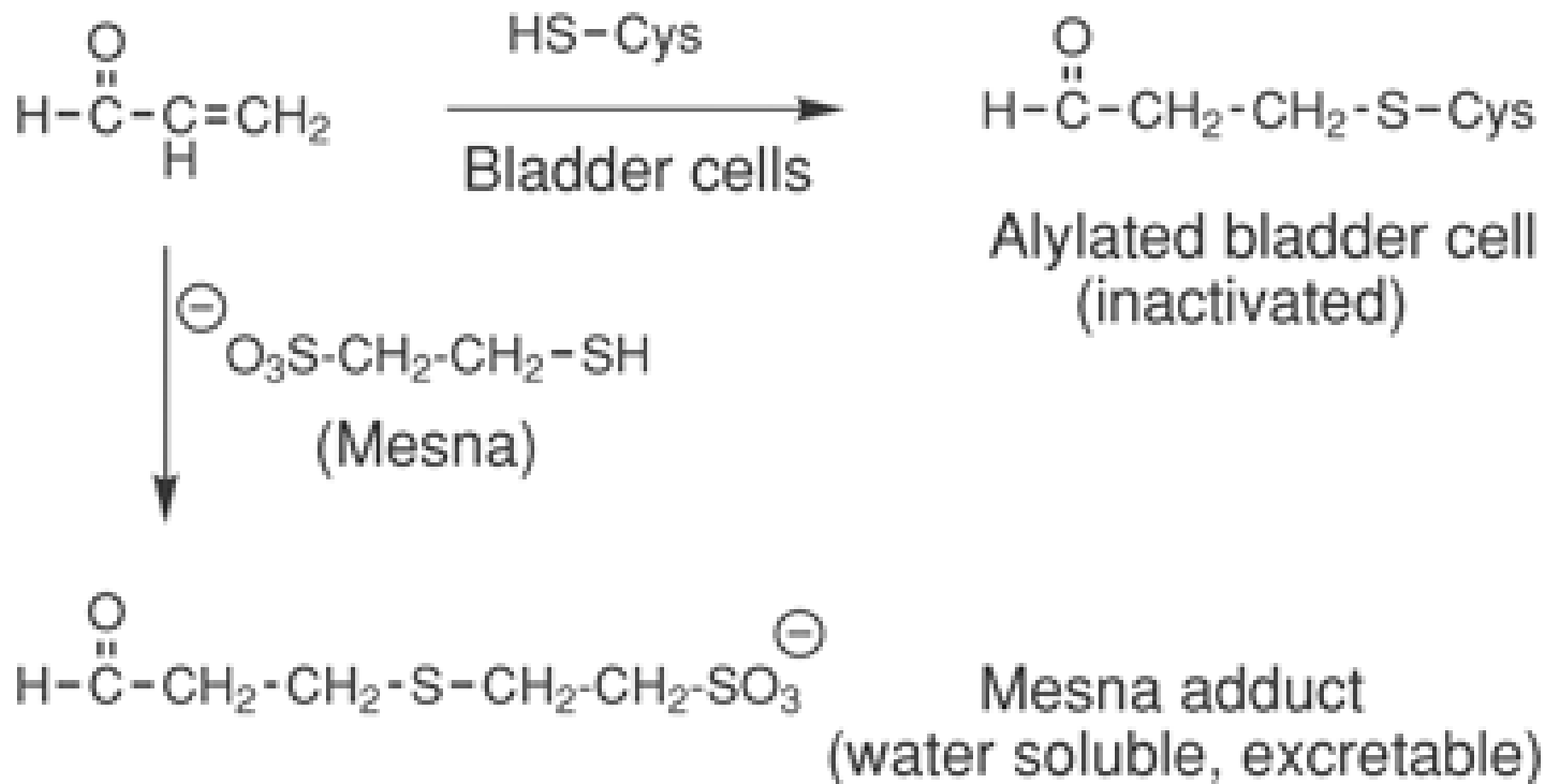
- It is biologically active *in vivo*;
 - It is converted to the active compound mainly in the liver.



- Acrolein causes urotoxicity, therefore Mesna which is acrolein antidote should be given $\text{HSCH}_2\text{CH}_2\text{SO}_3\text{Na}$ (Sodium mercaptoethane-sulfonic acid).
- **Mesna** is water soluble and forms a non-toxic product with acrolein.



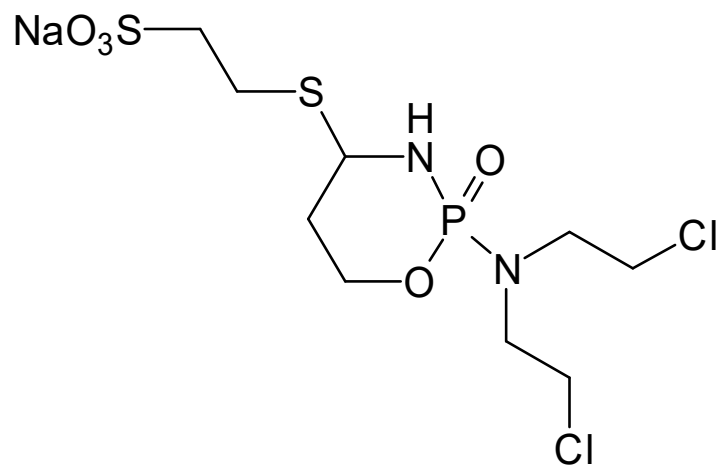
Antidotal role of mesna to acrolein



- Cyclophosphamide is used in the treatment of various malignant diseases including lymphoma, myeloma, and a variety of solid tumors.
- Adverse effects include myelosuppression and hemorrhagic cystitis (which may be prevented with mesna) as well as GIT, pulmonary, and cardiac toxicity, alopecia and sterility.
- It is administered as tablets.

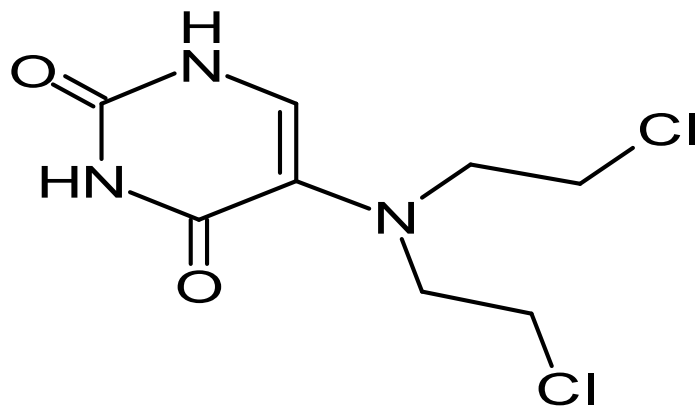
vi. Mafosamide

- It is a combination between cyclophosphamide and mesna



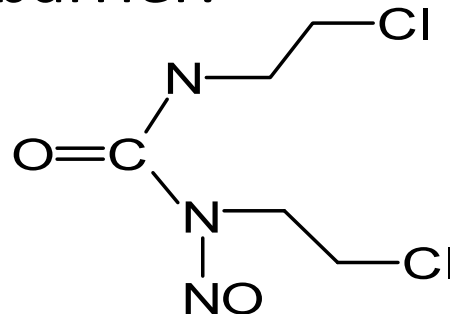
vi. Uracil 5-[Bis(2-Chloroethyl) amino] uracil.

- It is a combination of the structural features of N-mustard and a Nucleic acid bases which acts as a carrier for the active species.
- It is administered orally.

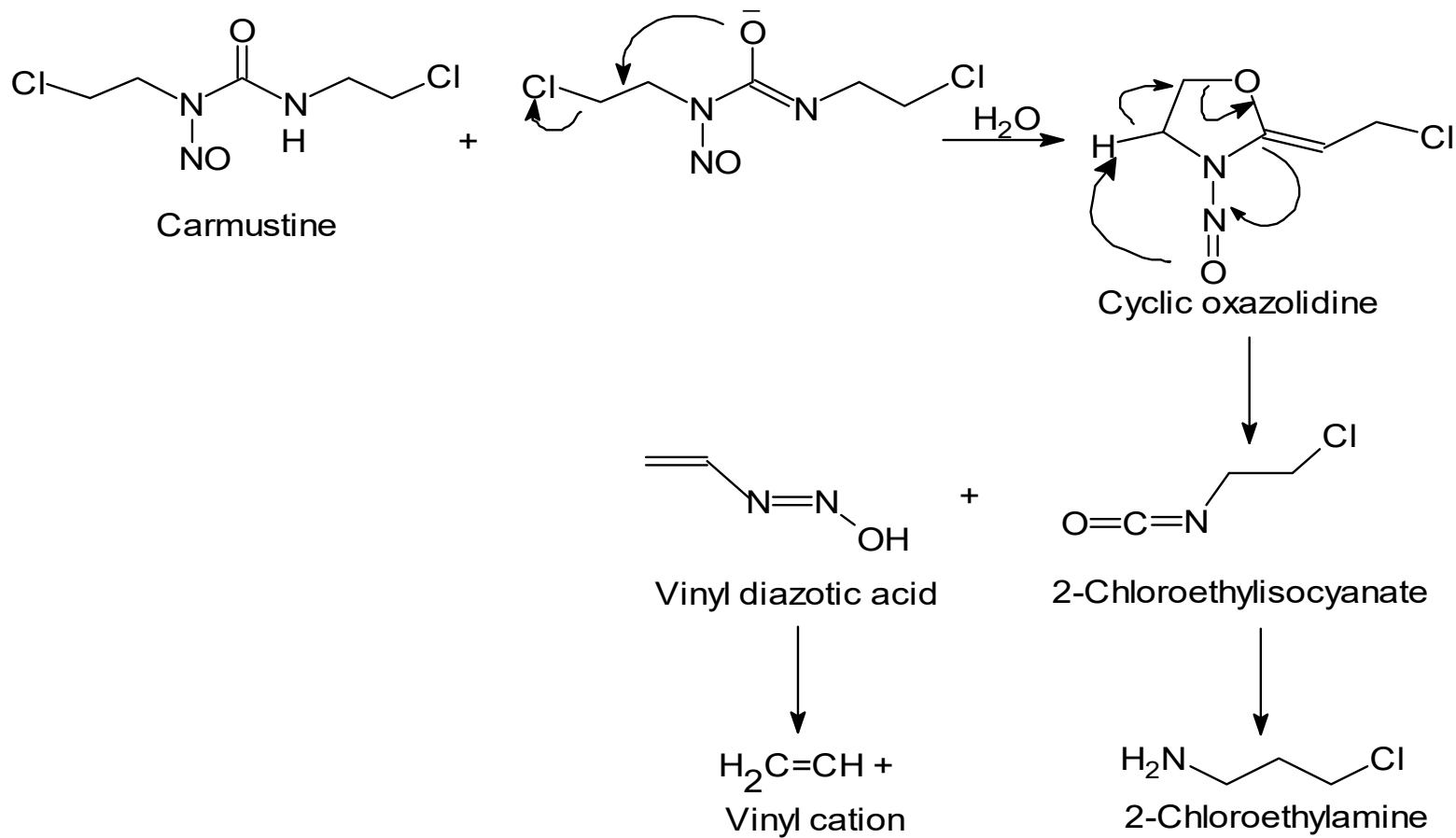


B. Nitrosoureas

- **i. Carmustine (BCNU) 1,3-Bis(Chloroethyl)-1-nitrosourea**
 - Carmustine is a neutral molecule that is highly lipophilic and poorly soluble in water.
 - These properties allow for efficient crossing of blood-brain barrier.

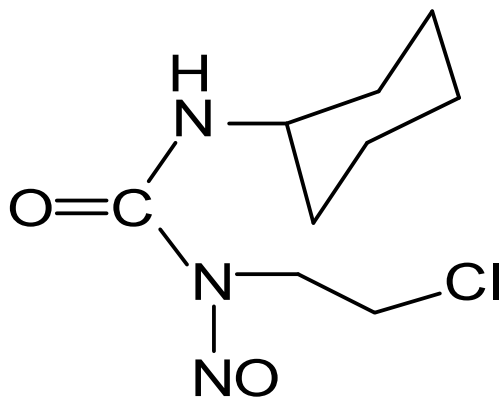


- Thought to alkylate DNA and RNA as a mechanism of action,
- It is not cross resistant with nitrogen mustards and alternate mechanisms of action involving enzyme inhibition by **carbamylation of proteins** have been suggested.
- It is rapidly metabolized after intravenous administration to form metabolites that have antineoplastic activity.



- Carmustine is used in the treatment of leukemia, lymphomas, malignant melanoma and brain tumors.
- Adverse effects include bone-marrow depression, pulmonary fibrosis, and effects on liver, kidneys, and eyes.

- **ii. Lomustine (CCNU) 1-(2-Chloroethyl)-3-cyclohexyl-1-nitrosourea**
 - It is a nitrosourea antineoplastic agent similar to carmustine.
 - It is formulated as capsules.

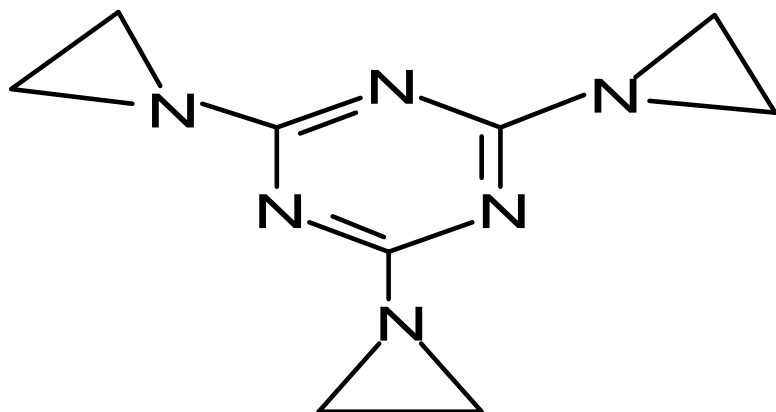


C. Aziridines (Ethyleneimines)

- Aziridine is a 3 membered nitrogen heterocycle that reacts with nucleophiles in order to relief ring strain.
- At acidic pH, the aziridine group is protonated to provide a reactive aziridinium ion that is known to alkylate DNA.

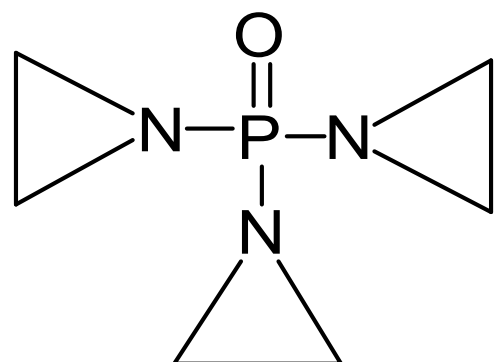
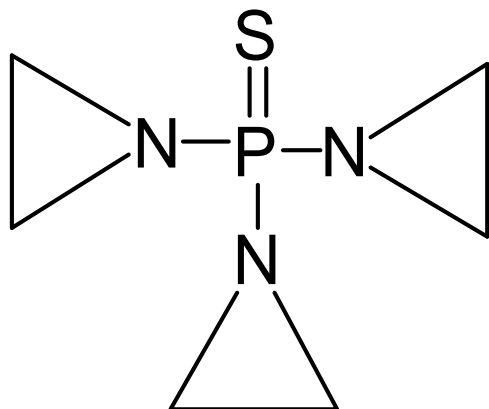
i. Triethylenemelamine (TEM, Tetramine) 2,4,6-Tris(1-aziridinyl)-s-triazine

- It is an aziridine containing compound.



- **ii. Thiotepa-Tris(1-aziridinyl)phosphine sulfide**

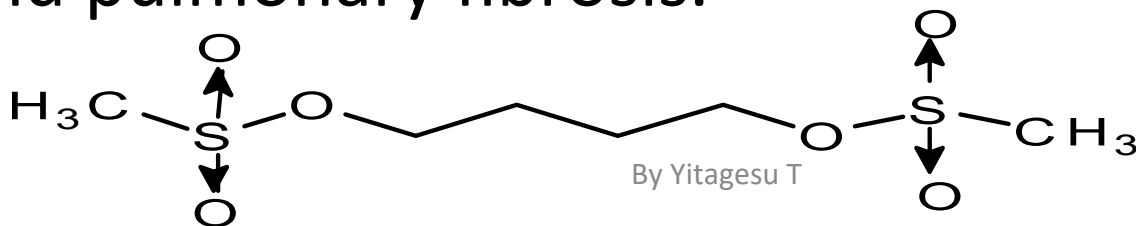
- An aziridine containing drug, used in carcinoma of breast, bladder, Hodgkin's and non Hodgkin's lymphoma.
- It is administered intravenously, intracavitary and intravesical.
- Metabolic desulfuration leads to a toxic metabolite.



Toxic metabolite

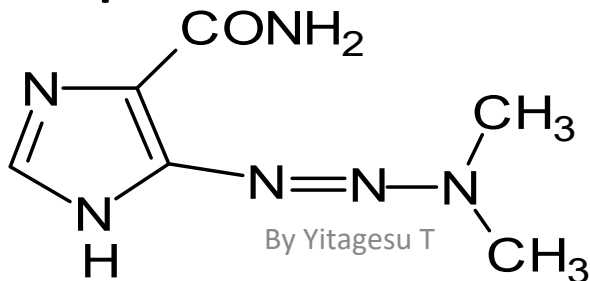
D. Sulphonic Acid Esters

- Busulphan (Myleran) 1,4-Butanediol dimethylsulfonate
- Busulphan is an alkylating agent with a methanesulfonic acid acting as a leaving group.
- It is used to treat chronic myelocytic leukemia.
- It is formulated as tablets.
- Adverse effects include prolonged and sometimes irreversible bone-marrow depression, and pulmonary fibrosis.



E. Triazines

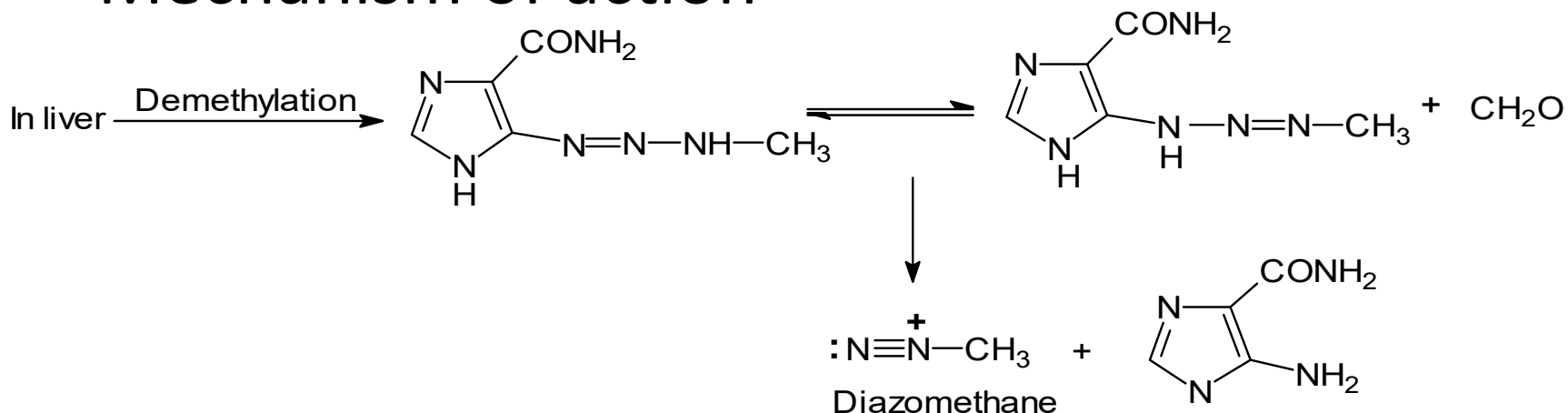
- Dacarbazine (DTIC) 5-(3,3-Dimethyl-1-triazenyl)imidazol-4-carboxamide
- Dacarbazine is a cell-cycle nonspecific antineoplastic agent which may function as an alkylating agent after its activation in the liver although other hypotheses, including inhibition of DNA synthesis by acting as a purine analogue, and interaction with sulfhydryl groups, have been proposed for its activity.



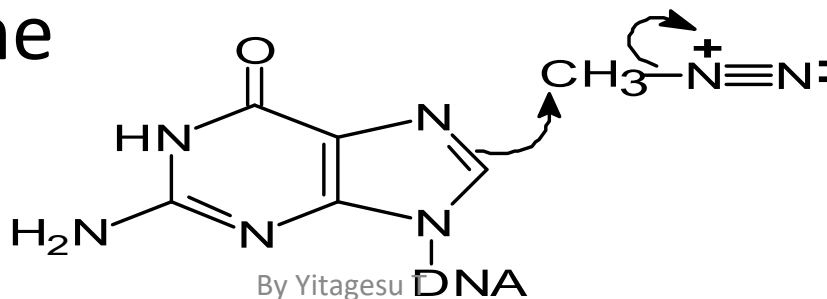
By Yitagesu T

- It is used mainly in the treatment of metastatic malignant melanoma and Hodgkin's lymphoma.
- It is administered intravenously and intramuscularly

- Mechanism of action

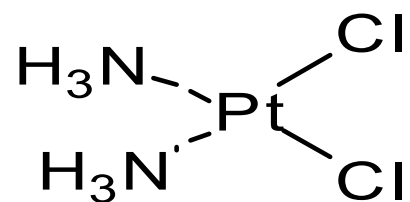


- Diazomethane is capable of methylating the N7 of guanine



F. Platinum compounds

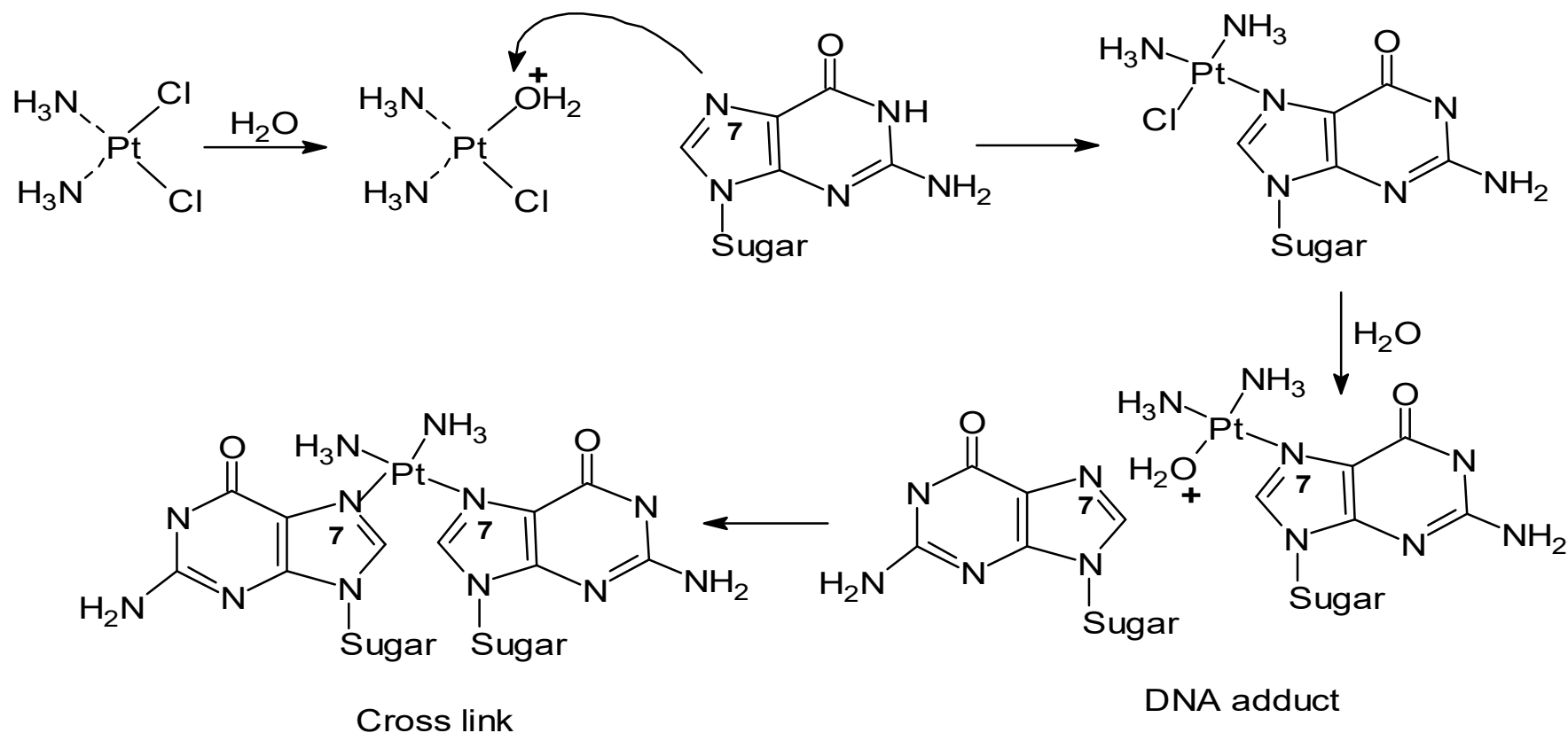
- Cisplatin (Platinol)



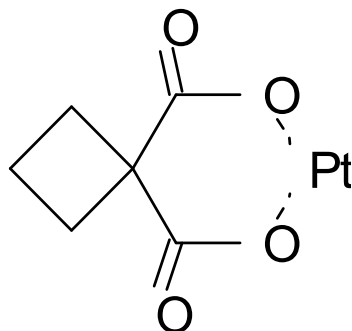
- It is a platin complex containing two ammonia molecules and two chlorine atoms in a cis configuration.
- It reacts with nitrogen atoms of DNA preferentially with N7 of deoxyguanylic acid forming intrastrand and interstrand cross-links.
- It is active against testicular cancer, lymphoma, carcinoma of the head and neck, ovarian cancer and bladder cancer, However it induces renal and neurotoxicities.

- It is administered intravenously.
- **Transplatin** is inactive as it lacks the stereochemistry required for cross linking.

Mechanism of Action



- Carboplatin (Paraplatin)



- It is less toxic than cisplatin and used in ovarian cancer.

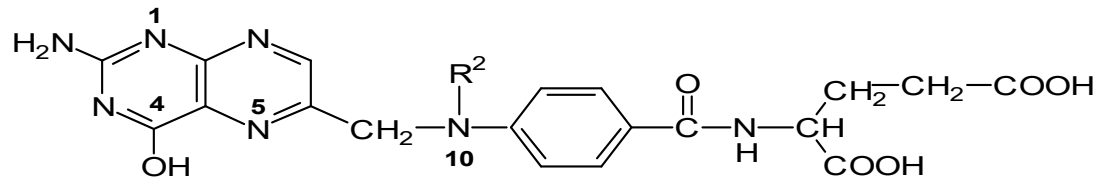
2. Antimetabolites

- Antimetabolites are compounds that prevent the biosynthesis or utilization of normal cellular metabolites.
- They are closely related in structure to the metabolite being antagonized.
- Many antimetabolites are **enzyme inhibitors**.
- They may combine with the active site as if they are the substrate or cofactor.

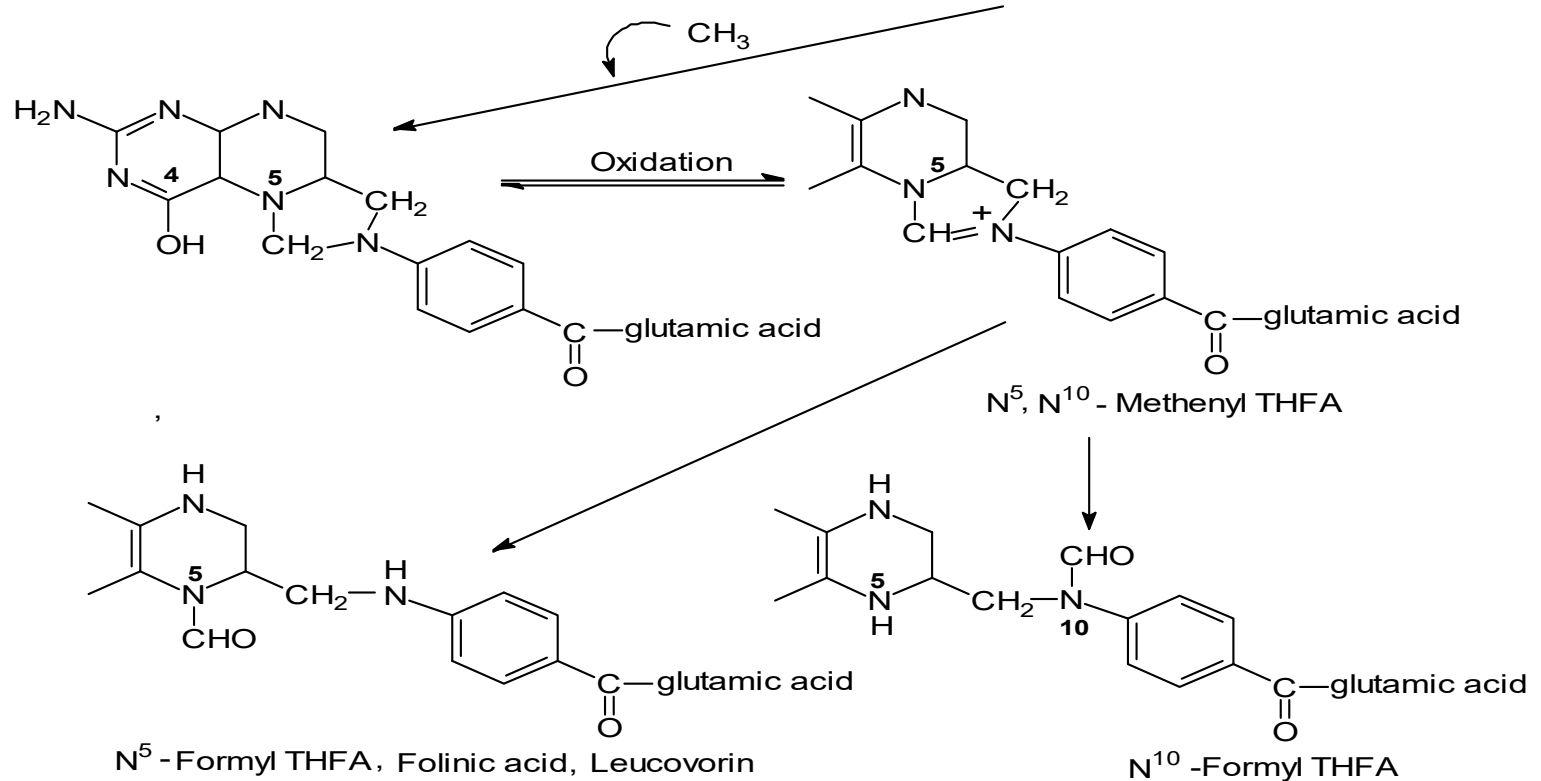
A. Folic acid Antagonists

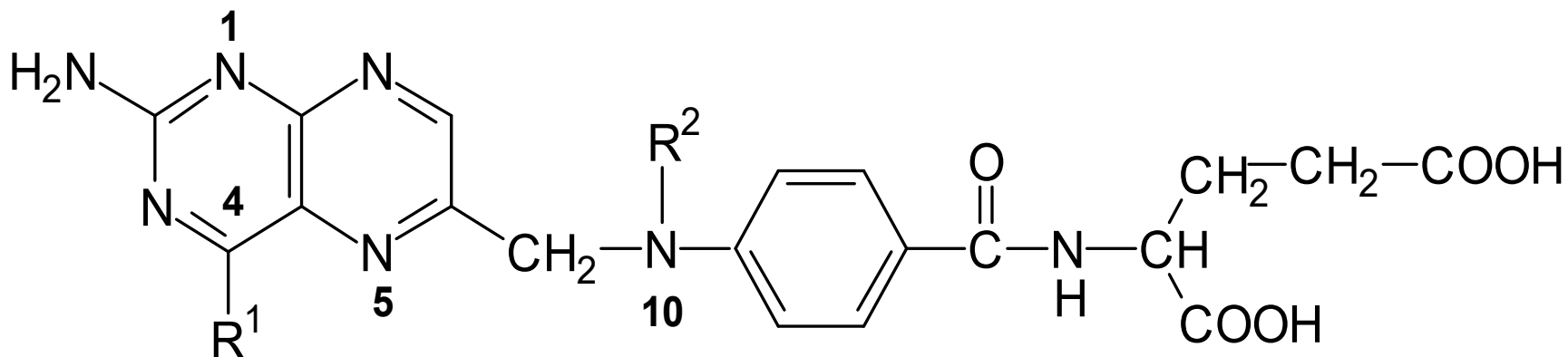
- **Folic acid** is essential for the synthesis of **nucleic acids**.
- It is reduced to **dihydrofolic acid** (DHFA) then to **tetrahydrofolic acid** (THFA) by **folate reductase**.
- **Tetrahydrofolic acid** picks up a **methyl** group from serine or methionine and is converted into **N5,N10-methylene tetrahydrofolic acid**.
- This compound is oxidized to **N5,N10-methenyl tetrahydrofolic acid**.
- Breaking of the covalent bond between the formyl carbon atom and either N5 or N10 results in N10-formyl or N5-formyl tetrahydrofolic acid (folinic acid, leucovorin), respectively.

- In this loosely bound N5,N10-bridged position the one-carbon formyl group is transferred to other compounds that are then utilized in various essential pathways.
- It finds its way into C2 and C8 of purine bases, the C5 of deoxyuridilic acid and the methyl group of thymine.
- Tetrahydrofolic acid is also involved in biologically important reactions not directly involved in cell proliferation.



Folic acid



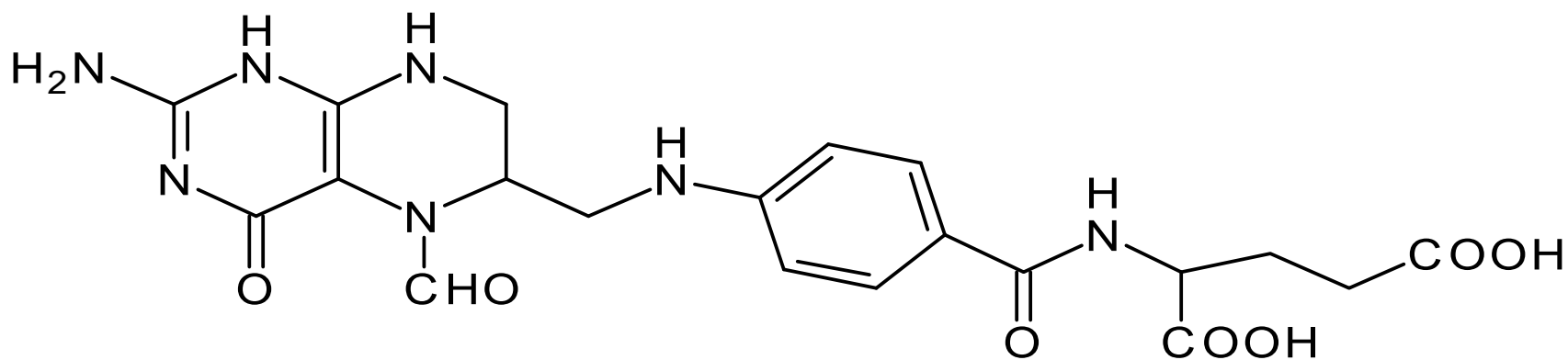


Folic acid	OH	H	Metabolite
Aminopterin	NH ₂	H	Antimetabolite
Methotrexate	NH ₂	CH ₃	Antimetabolite

- Methotrexate is 4-amino-N10-methyl pteroylglutamic acid.
- Methotrexate acts as an antifolate by binding almost **irreversibly to the enzyme dihydrofolate reductase and preventing the formation of the co-enzyme** tetrahydrofolic acid, essential for DNA synthesis and for replication of animal cell.
- **Methotrexate inhibits not only the enzyme DHFR, but also thymidylate synthetase, these processes are vital for the metabolism of nucleic acid in the cells.**
- Methotrexate inhibits folate reductase at two steps.
- The drug binds much more strongly to the enzyme than does folic acid.

- Folic acid therefore cannot reverse the toxic effects of the drug which accounts for the relatively long lasting effects of methotrexate.
- Folinic acid, unlike folic acid can compete with methotrxate since it can still carry out one-carbon in spite of the fact that folate reductase is inhibited.
- This is the rationale for using folinic acid (leucovorin) as an antidote for methotrexate.
- Methotrexate is administered in tablet form and used to treat acute leukemia, meningeal leukemia, carcinoma of breast, head, neck, lung, ovarian and bladder carcinoma.

- Leucovorin Rescue

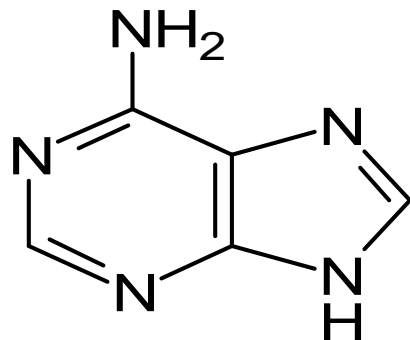


- 24 Hours infusion of methotrexate is more toxic than one push injection of the same amount. So giving the patient about 200 times the therapeutic dose will allow cancer cells to be exposed rapidly to very high doses in short time.
- Injection of leucovorin after few hours will supply normal cells with THFA despite inhibition of folate reductase by methtroxate.
- The result is a short exposure time of cancer cells to high doses of methotrexate and much more favorable therapeutic index.

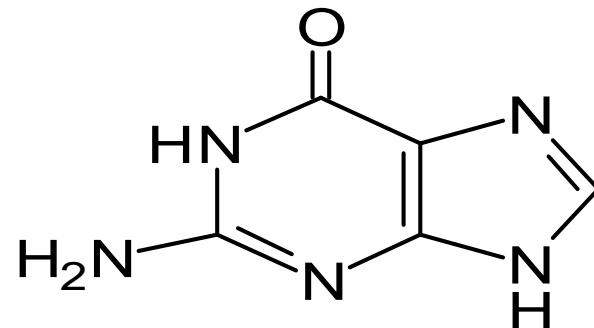
B. Purine Antagonists

- 6-Mercaptopurine (6MP) and 6-thioguanine (6TG) are examples of purine antimetabolites.
- Both are not active until they are converted to their respective nucleotides by a **pyrophosphorylase enzyme** in the cell.
- Since this is an intrinsically destructive process to the cell, it is referred to as a lethal synthesis.
- Tumor cells or normal cells that have very low levels of this enzyme are resistant to these two agents.

- Absence of this enzyme function is not lethal to the cell since the synthesis of purines in the cell occurs by different pathways.
- They interfere with the de novo synthesis of purines.



Adenine

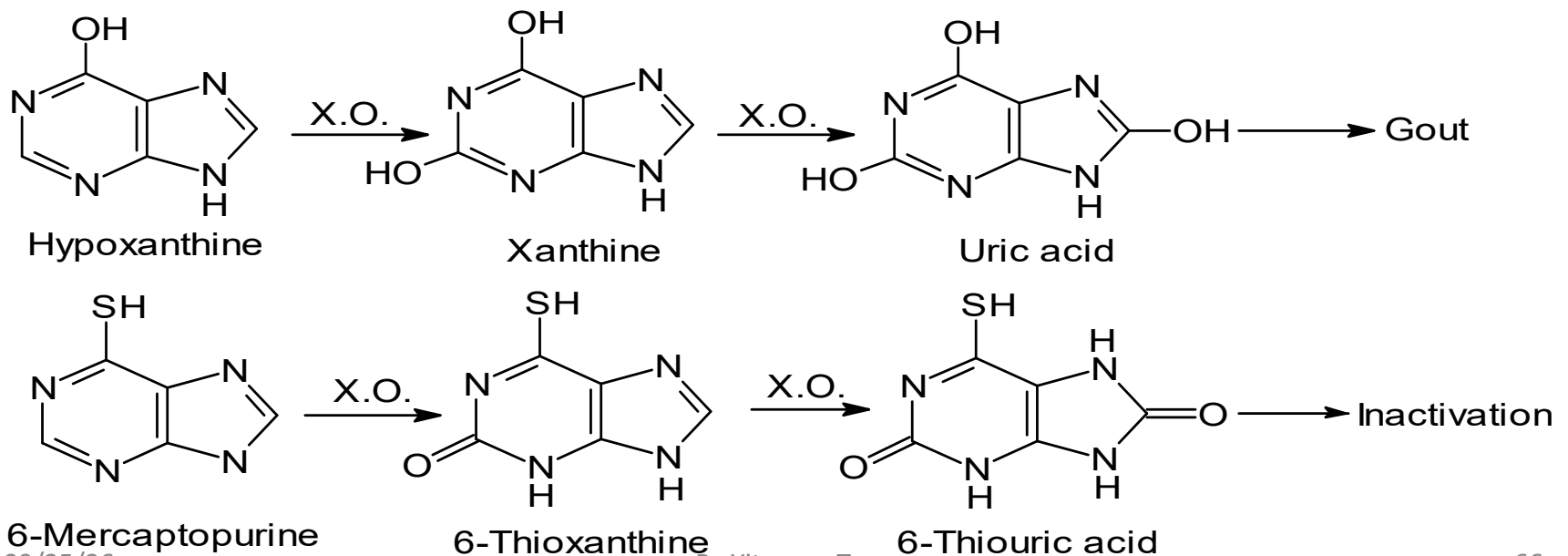


Guanine

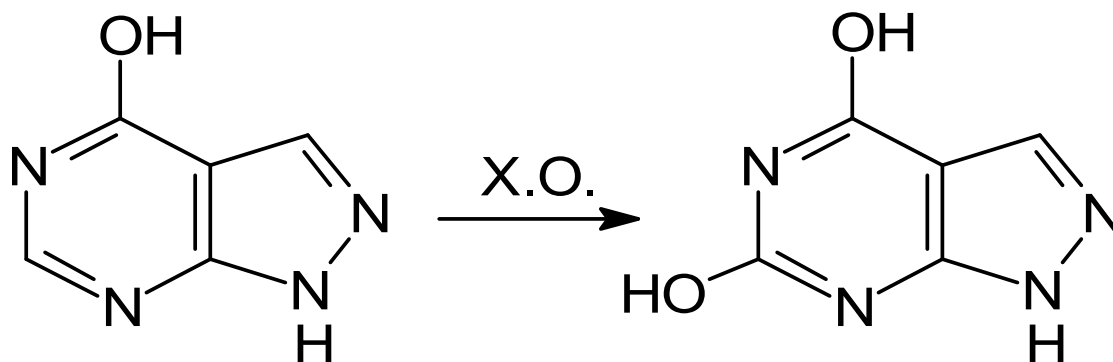
Natural bases

- Co-Administration of Allopurinol with 6-Mercaptopurine
- 6MP is metabolized by xanthine oxidase (XO) to 6-thioxanthine then to 6-thiouric acid, while 6TG is not affected by this enzyme.
- On the other hand, XO is responsible for the oxidation of hypoxanthine to uric acid which deposits in joints leading to gout.
- Allopurinol is a structural isomer to hypoxanthine and is a xanthine oxidase inhibitor but is not uricosuric acid.
- When allopurinol is given with 6MP, it will prevent the metabolic effect of the enzyme on 6MP, consequently the normal dose of 6MP will be toxic.

- Therefore, in case of administering 6MP with allopurinol the dose must be reduced; decreasing the risks of nephrotoxicity and acute gout formation.



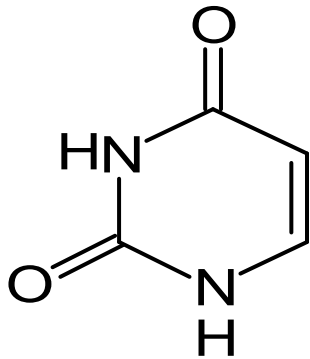
- The main importance of allopurinol as an adjuvant to chemotherapy is that it prevents uric acid toxicity caused by the release of purines from destroyed cancer cells.



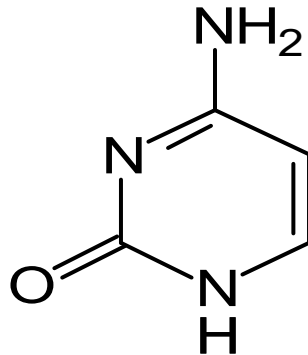
Allopurinol

Alloxanthine

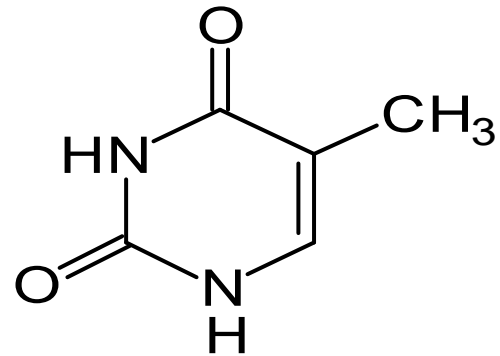
C. Pyrimidine Antagonists



Uracil



Cytosine

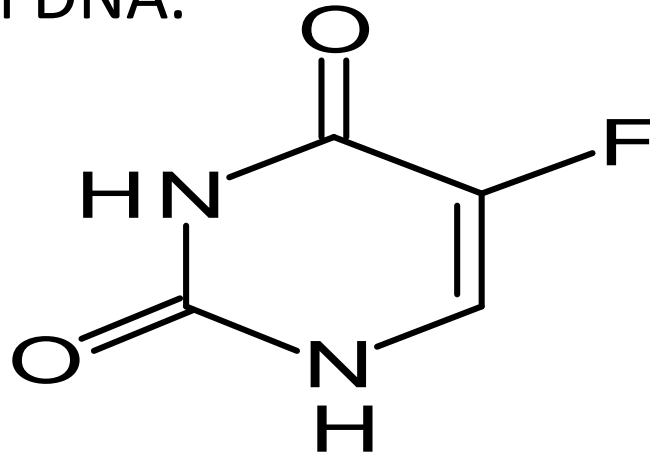


Thymine

└── In RNA ──┐
└── In DNA ──┘

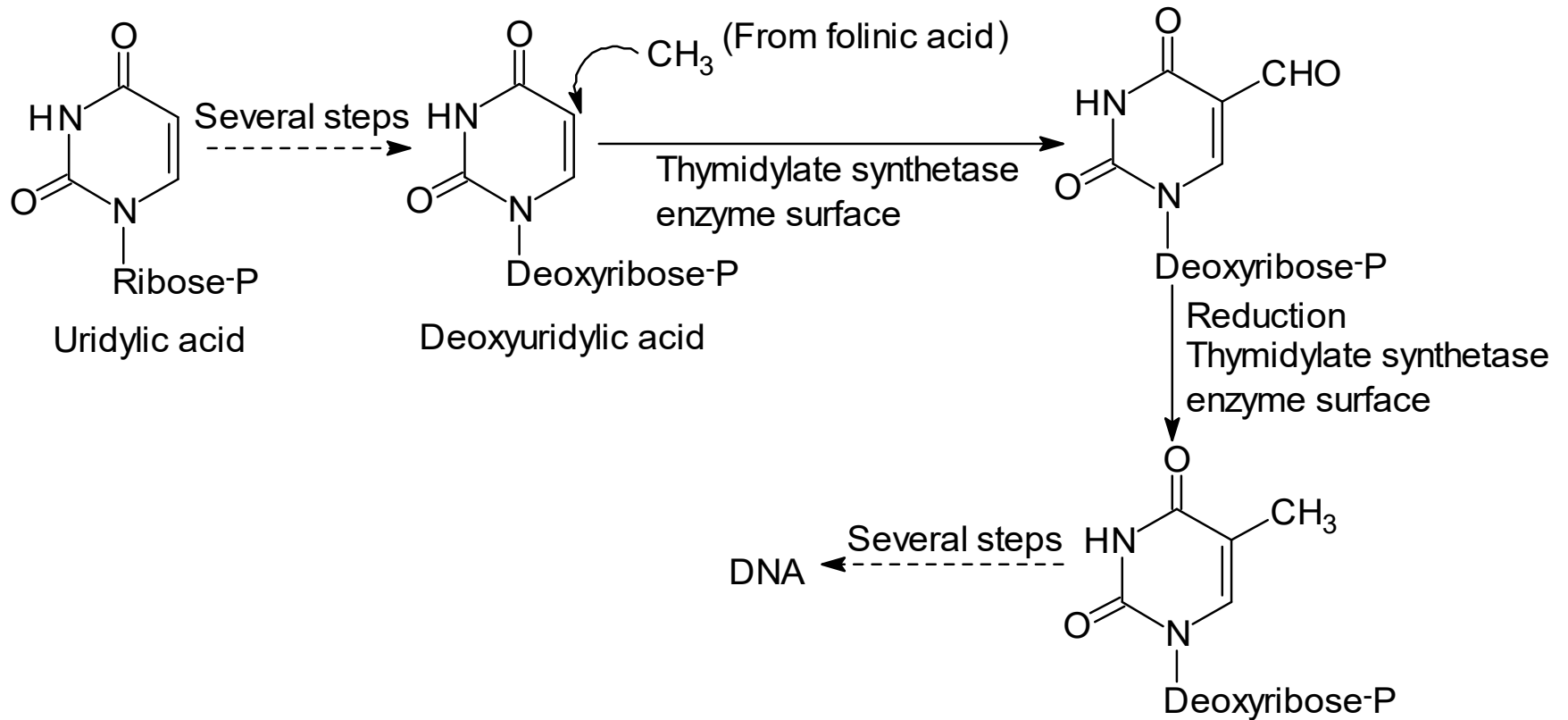
Natural pyrimidines

- 5-Fluorouracil (5FU)
 - Uridylic acid, the uracil nucleotide, goes through several steps to form deoxyuridylic acid which
 - when methylated at position 5 becomes deoxythymidylic acid, the pyrimidine nucleotide found only in DNA.



- The enzyme that carries out this transfer and reduction of a CH₃ group from the reduced folates is **thymidylate synthetase**.
- Thus a very close analog of uracil or its nucleotide would be able to bind to the reactive site of the enzyme and **prevent the methylating step**.
- The molecular modification that affected this was the substitution of the hydrogen in position 5 of uracil with a fluorine atom.

- *This substitution accomplishes two things:*
 1. Since fluorine is only slightly larger than the hydrogen atom, very little molecular perturbation occurred resulting in the enzyme not recognizing the difference and accepting it instead of the natural substrate, in fact, the affinity for the drug was greater. H= 1.2 Å, F= 1.33 Å Cl= 1.73 Å, Br= 1.95 Å and CH₃= 2 Å. Cl and Br are close to CH₃ leading to bypass, so no enzyme blockade.
 2. Since position 5 is occupied, it cannot be methylated.
- Thus inhibition of thymidylate synthetase by 5-fluorouracil is the major reason for its cytotoxicity

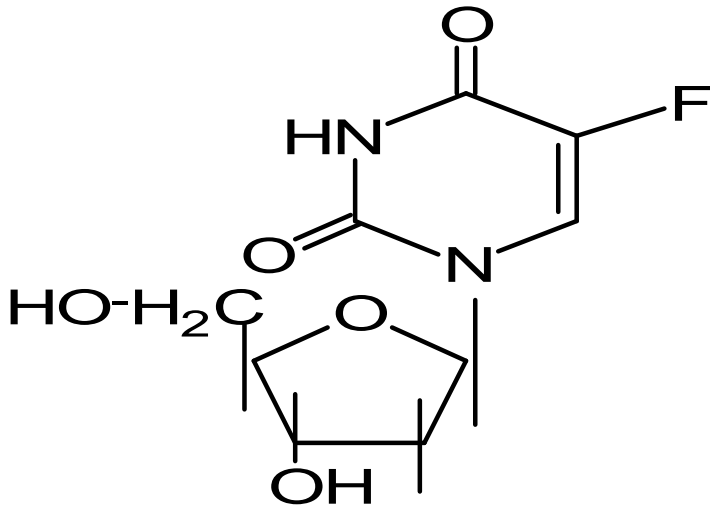


- Actually 5-fluorouracil is not intrinsically toxic.
- It only becomes effective after the metabolic steps outlined in the above scheme.
- It is administered intravenously for the treatment of carcinoma of breast, colon, rectal, stomach, ovarian, cervical, prostate, bladder, head and neck carcinoma.
- It is administered topically for the treatment of skin and actinic keratoses.

ii. Floxuridine (5-Fluorodeoxyuridine)

- 5- Fluorouracil-2'-deoxyribose,
- It is easier to be activated than 5FU since it **needs only the phosphate group to be added.**
- Cancerous cells may undergo mutation and develop resistance upon prolonged treatment with purine and pyrimidine antimetabolites.

- (5-Fluorodeoxyuridine)



3. Hormones and Antihormones

- Tumors derived from hormone-sensitive tissues may be hormone dependent.

Their growth can be inhibited by

- hormones of opposing actions,
 - hormone antagonists or
 - agents that inhibit the synthesis of the relevant hormone.
- ❖ Hormones essential for reproduction are also responsible for development and growth of breast, prostate and uterine cancer.

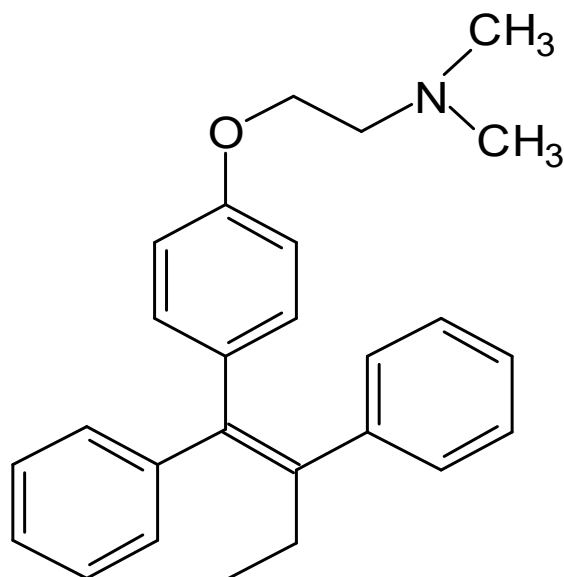
A. Hormone-Dependent Breast Cancer

- Estrogens bind to an estrogen receptor in estrogen target site (uterus, vagina, pituitary gland) to give E-R complex with subsequent response.
- The more is the cells in the tumor that contain estrogen; the higher is the estrogen receptor content.
- The tumor is said to be estrogen receptor rich (estrogen receptor positive).
- **This tumor type responds well to endocrine therapy (antiestrogen).**

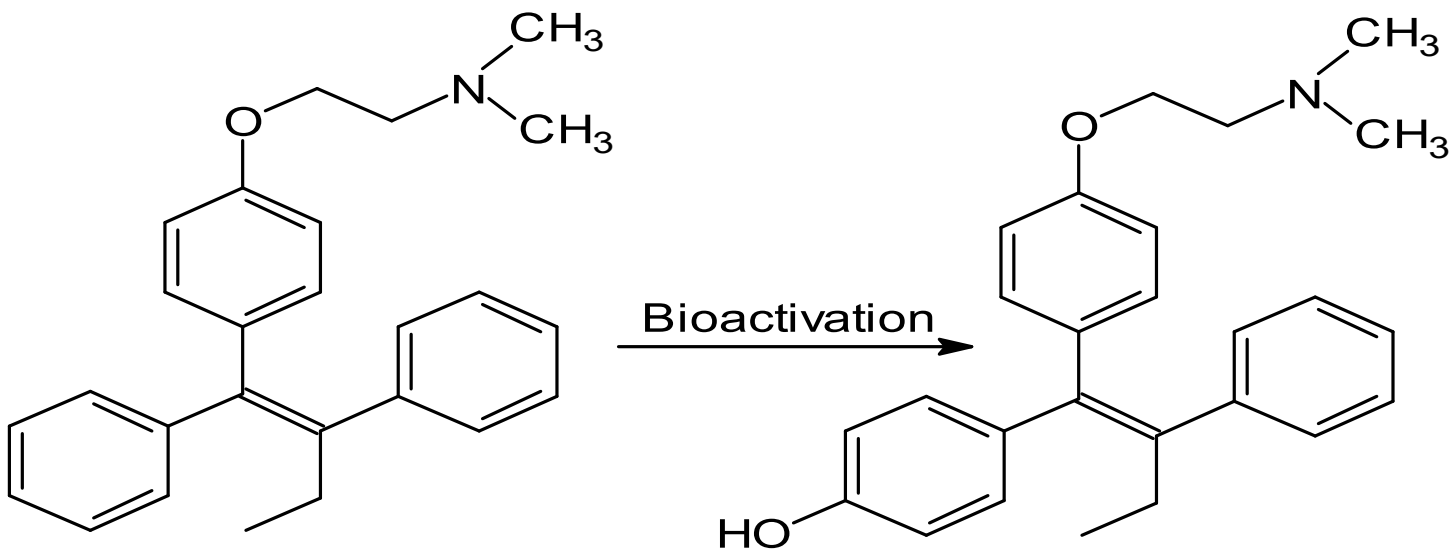
- On the other hand, tumor with low estrogen receptors is said to be estrogen poor (estrogen receptor negative) responds poorly to endocrine treatment.
- **Ethinylestradiol** is given orally in the treatment of breast cancer in post menopausal women.

- **Androgens** are active against metastatic breast cancer about 20% of postmenopausal women.
- **Progesterone** and its analogs are active against certain neoplasms that are stimulated by estrogens.
- High doses of progestins or androgens antagonize the proliferative action of estrogen.
- Endometrial carcinoma is treated with estrogens

- i. Tamoxifen 2-[4-(1,2-Diphenyl-1-butenyl)phenoxy]-N,N-dimethylethanamine



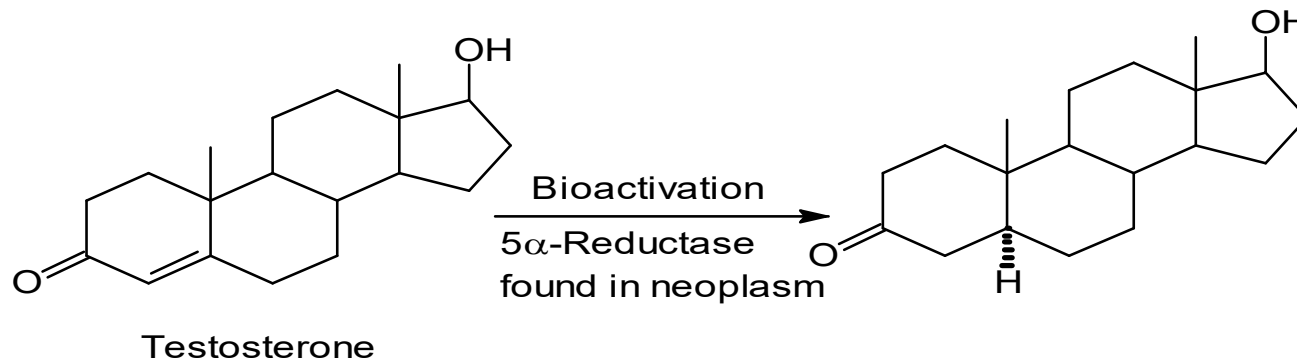
- It is a Nonsteroidal antiestrogen used in the treatment of the advanced breast cancer.
- In breast tissues, **tamoxifen competes with endogenous estrogen for the estrogen receptor and inhibits the transcription of estrogen-responsive genes.**
- It is bioactivated first to produce the active hydroxylated derivative.



B. Hormone-Dependent Prostatic Cancer

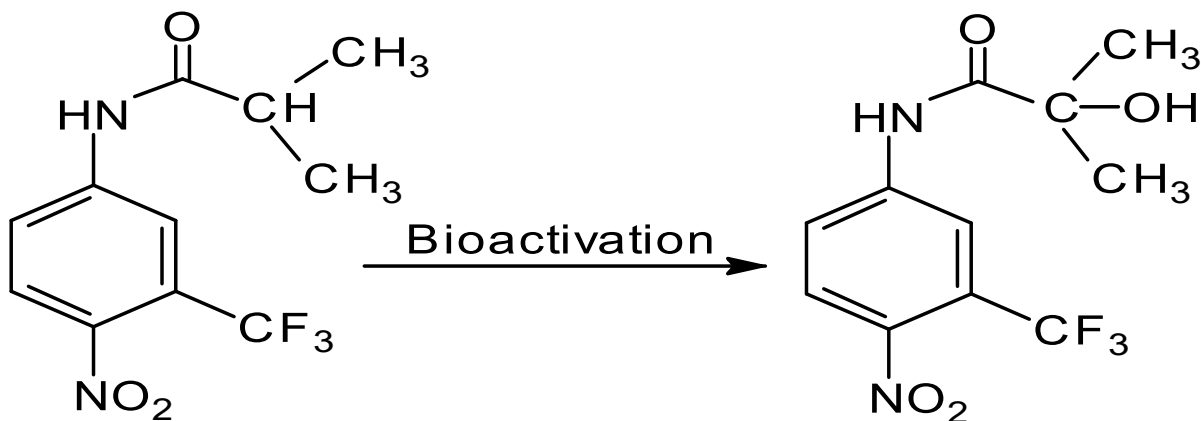
- Most of prostatic tumors are androgen-dependent.
- The action of androgens in prostate involves metabolic activation of circulatory androgens to a derivative with high binding affinity for androgen receptors.

- Androgen antagonists are used for the treatment of metastatic prostate cancer.
- They block the action of testosterone and dihydrotestosterone.



- *Antiandrogens may be classified into :*
 - a. Steroidal compounds as cyprotone , chlormadinone and megestrol.
 - b. Nonsteroidal compounds as flutamide (eulexin) which is bioactivated first

- i. Flutamide (Eulexin) 2-Methyl-N-[4-nitro-3-(trifluoromethyl)phenyl]propanamide
 - It is administered as capsules

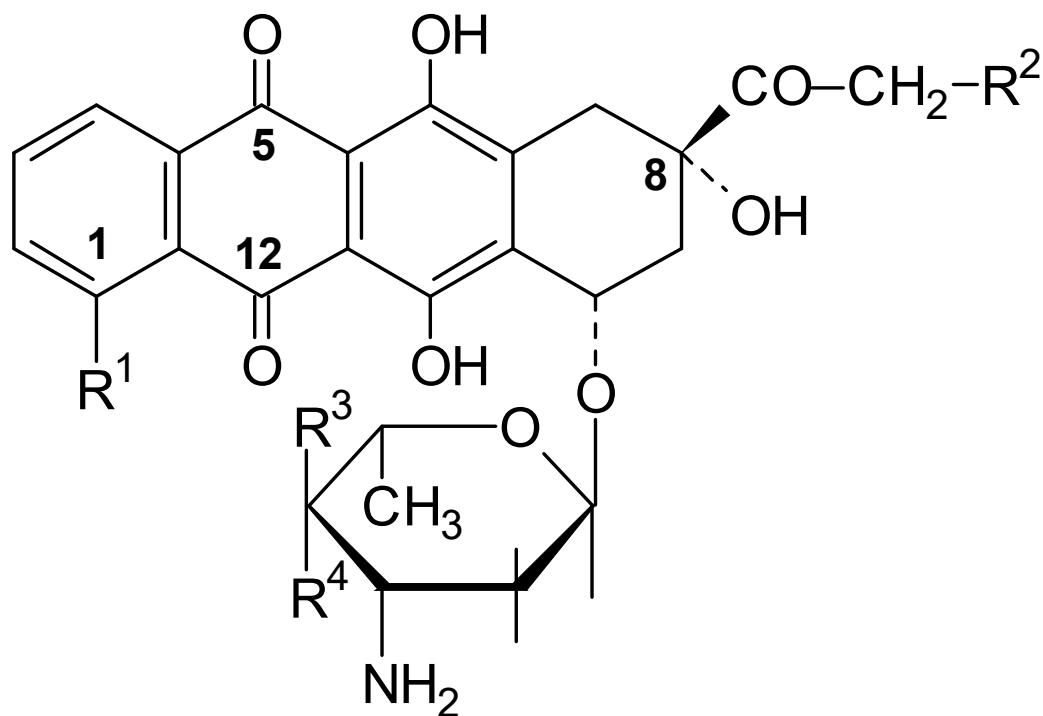


4. ANTIBIOTICS

- Several compounds that were originally evaluated for their antibiotic activity have been clinically useful anticancer agents.
- However, they were rejected as antibiotics because of their toxicity.
- This property was subsequently turned into asset with their application as anticancer agents.
- The source of most antitumor antibiotics is from microbial fermentation.

Anthracyclines

- These antibiotics represent a major class of antineoplastic agents.
- It is characterized by having a tetracyclic quinone containing ring nucleus to which is attached a unique daunosamine sugar.
- It was isolated from the fermentation broths of *Streptomyces peucetius*.
- Several semisynthetic derivatives have been prepared in an attempt to reduce the cardiotoxicity common to this class of compounds.
- These include daunorubicin, doxorubicin, idaurubicin and epirubicin.



Doxorubicin: $R^1 = \text{CH}_3\text{O}$, $R^2 = \text{OH}$, $R^3 = \text{H}$, $R^4 = \text{OH}$

Daunorubicin: $R^1 = \text{CH}_3\text{O}$, $R^2 = \text{H}$, $R^3 = \text{H}$, $R^4 = \text{OH}$

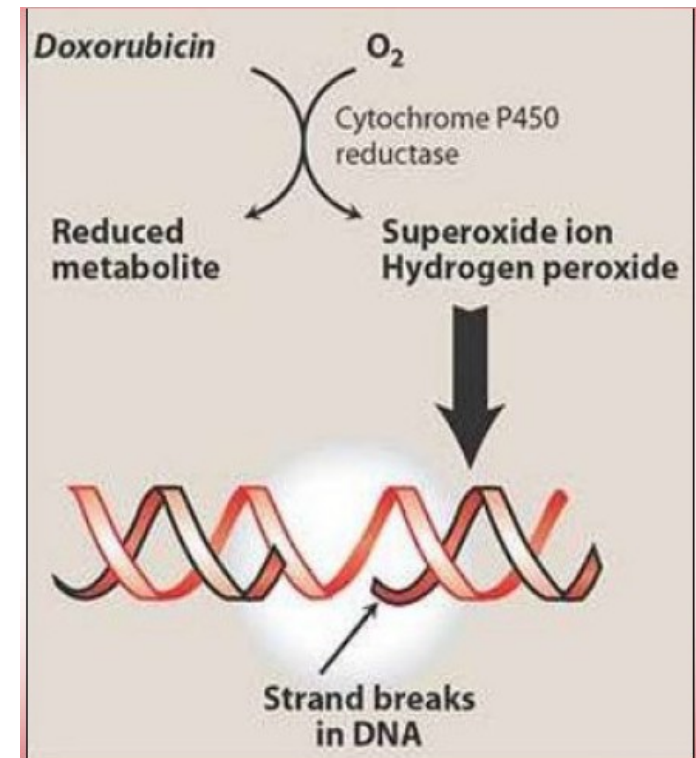
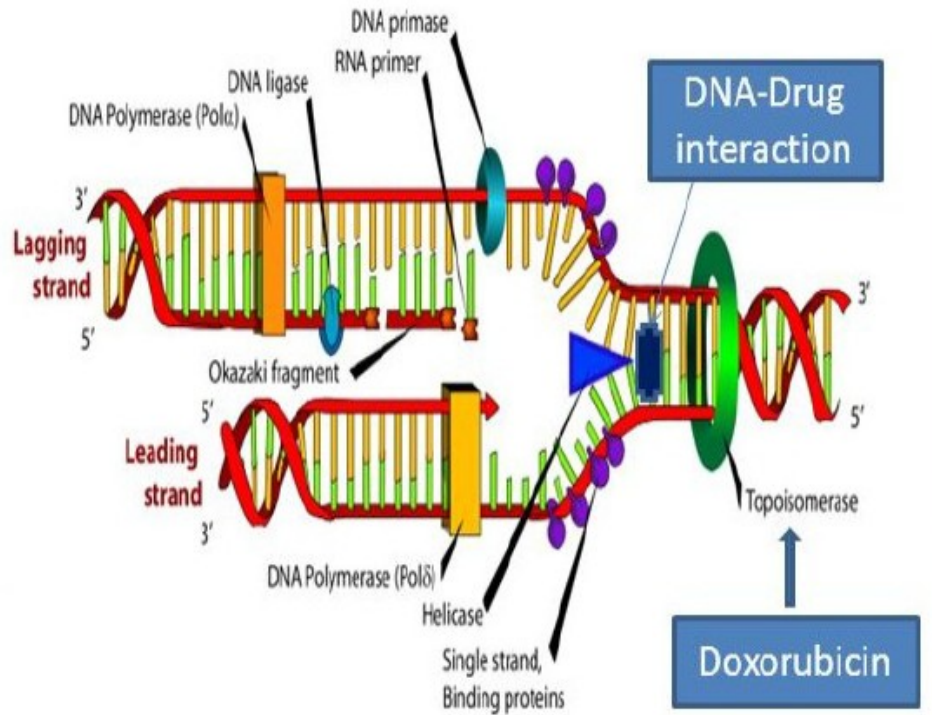
Idarubicin: $R^1 = R^2 = R^3 = \text{H}$, $R^4 = \text{OH}$

Epirubicin: $R^1 = \text{CH}_3\text{O}$, $R^2 = \text{OH}$, $R^3 = \text{OH}$, $R^4 = \text{OH}$

Mechanism of action

- The flat structure of anthraquinone nucleus results in the ability of anthracyclines to interact with DNA perpendicularly to its long axis.
- The amino sugar confers added stability to this binding through its interaction with the sugar phosphate backbone of DNA.
- The results of interaction can lead to single- and double-stranded DNA breaks.
- Because of the anthraquinone ring system, the anthracyclines are capable of generating reactive oxygen species such as .OH radical and super oxide radical anions(.O-O-).
- These free radicals produce destructive effects upon the cell.
- The generation of free radicals may also account for the cardiotoxicity of the anthracyclines.
 - Due to lack of oral activity, they are administered intravenously.

Mechanism of action



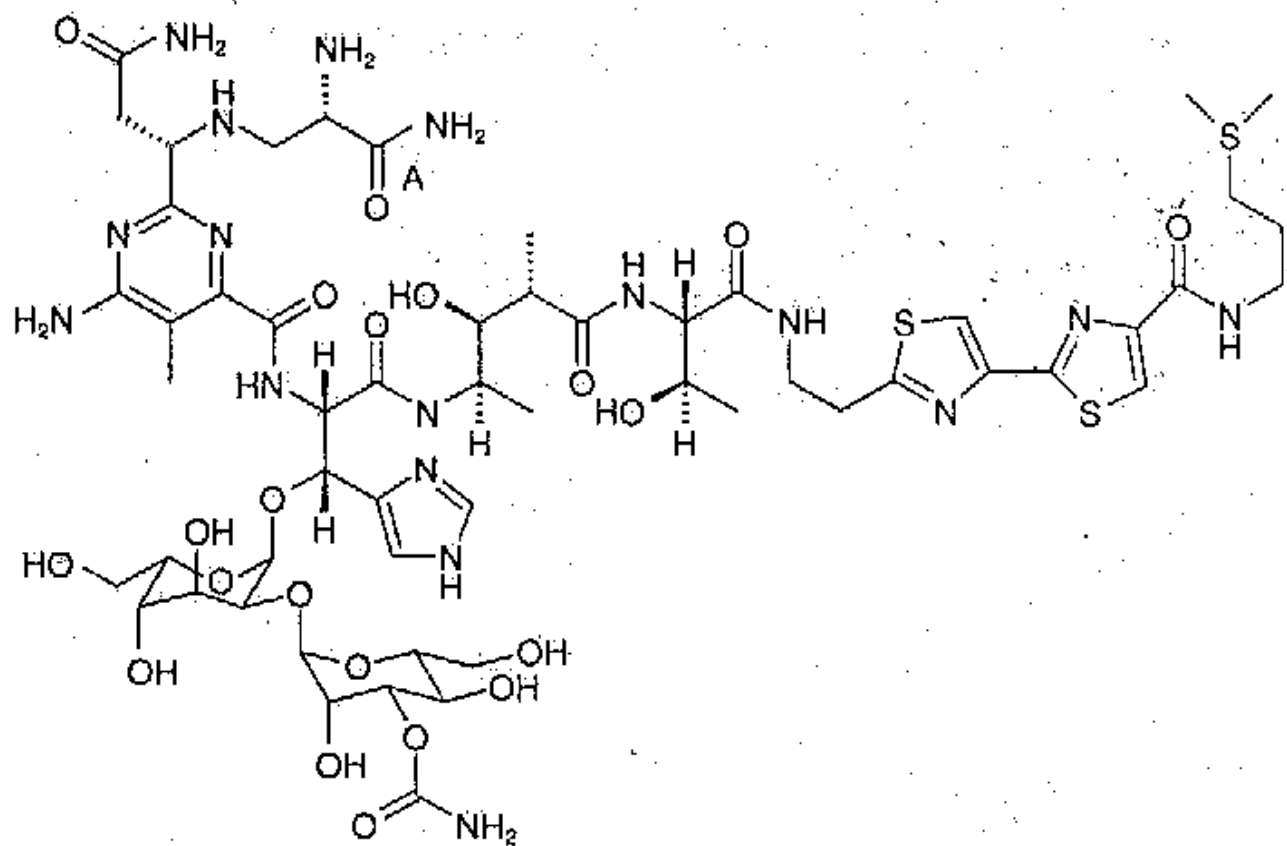
ii. Bleomycins

- Bleomycins are a group of metal-chelating glycopeptide antibiotics that degrade DNA; its molecule features two pivotal domains.
- One is the DNA domain, which allows interaction with DNA and RNA.
- **The second binding site is a metal binding domain which is involved in chelation of ferrous iron and interaction with the oxygen, resulting in the oxidation of the iron and generation of superoxide.**
- It is used for the treatment of carcinomas of the head and neck.

Metal binding
domain

DNA binding
domain

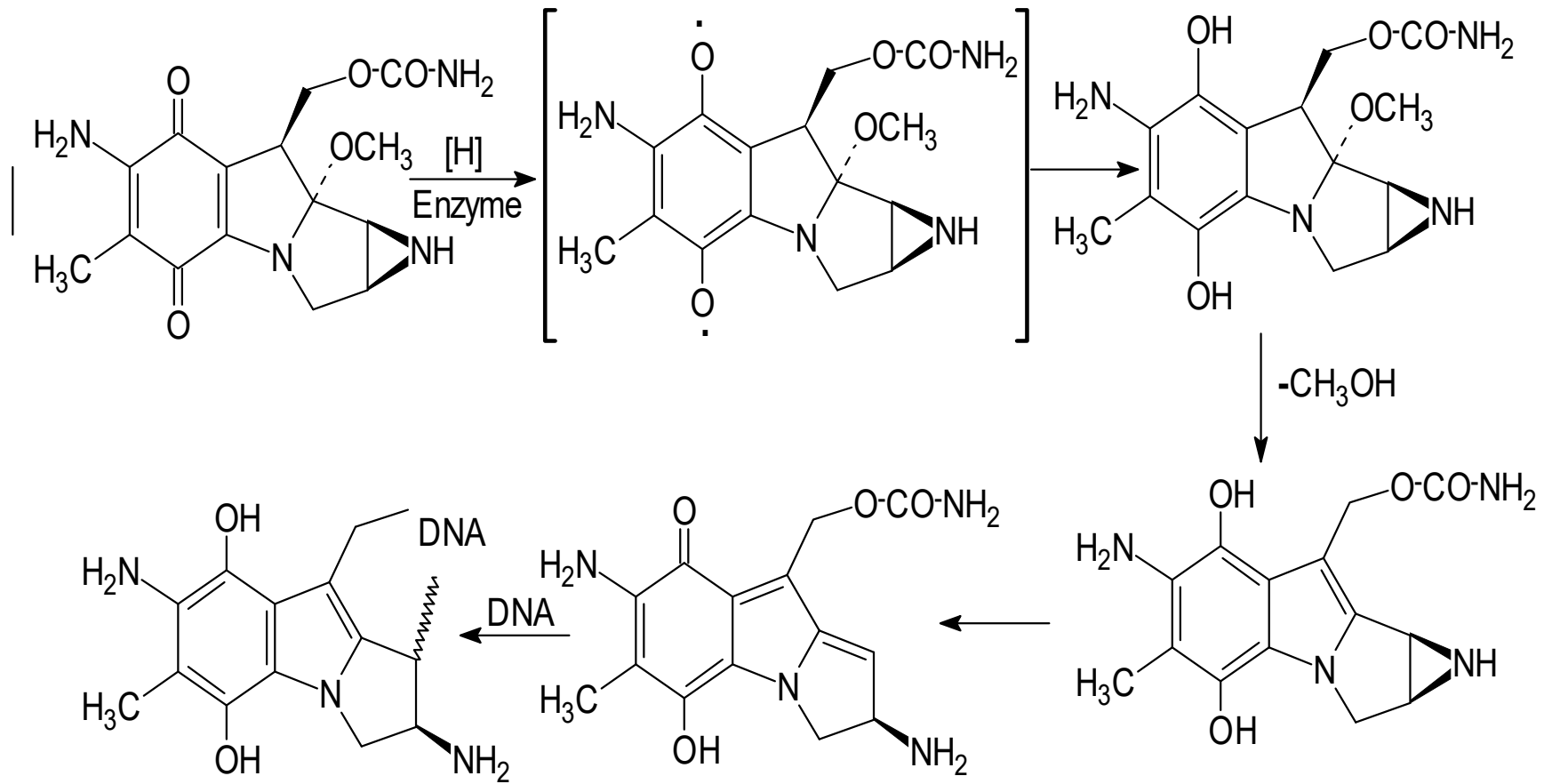
Cu(I), Cu(II)
Fe(II), Fe(III)
Mn(II), Mn(III)



iii. Mitomycin C

- It is an antitumor antibiotic isolated from *Streptomyces caespitosus*.
- It possesses several functional groups capable of antitumor activity.
- The **quinone moiety, the aziridine ring system** and **the carbamate functionality** are involved in the action of the drug and the drug is classified as a **bio-reductive alkylating agent**.
- After bio-reductive activation, followed by elimination of methanol, the compound is capable of reaction with nucleophiles in DNA leading to either mono- or dialkylation products.
- The drug is poorly absorbed via oral administration and is administered intravenously.

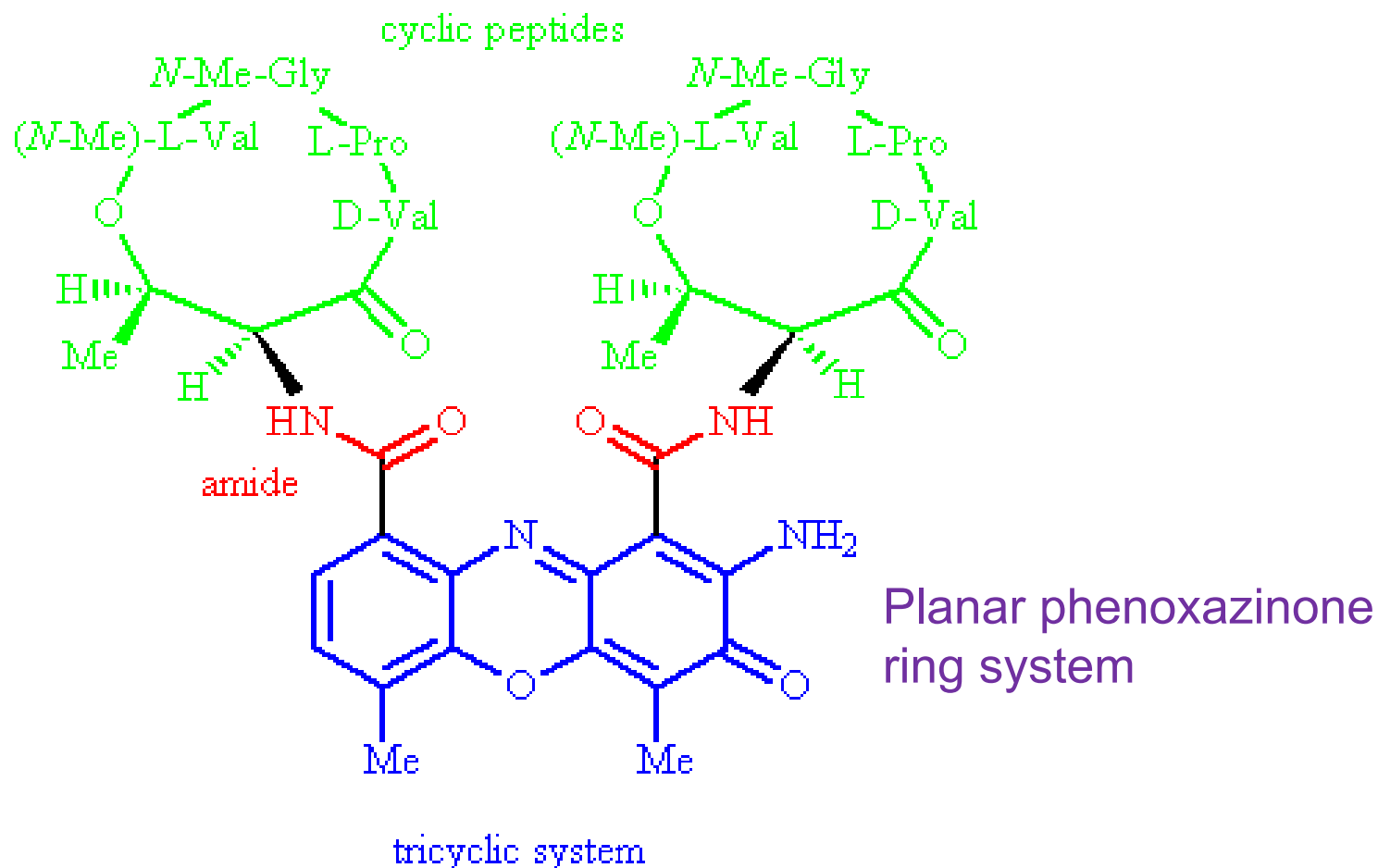
Mechanism of action



iv. Dactinomycin (Actinomycin D)

- Dactinomycin is the main antibiotic of *Streptomyces parvullus*.
- It consists of **tricyclic phenoxazone ring** and two identical pentapeptide lactones.
- The pentapeptides are made up of L-proline and L-threonine plus the essential amino acids D-valine, sarcosine and N-methylvaline.
- The ring system is planar and can intercalate or insert into DNA between base pairs.
- The distortion caused by the presence of the intercalating agent affects the action of topoisomerase II, which normally regulates unwinding of coiled double-stranded DNA.
- This in turn, interferes with DNA replication and transcription.

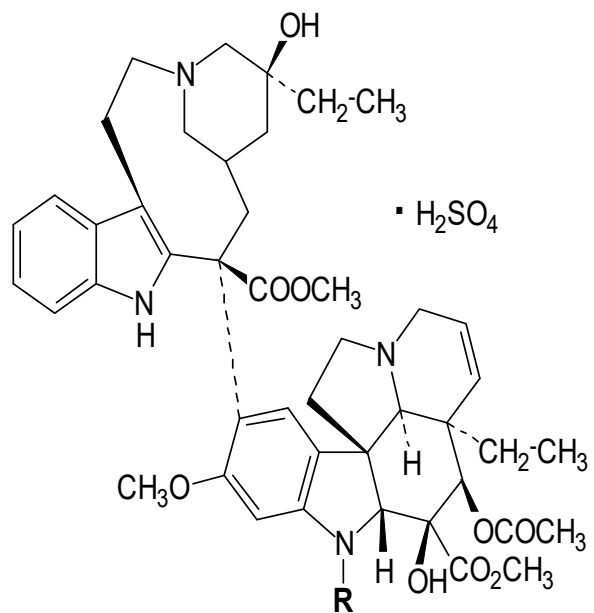
Dactinomycin



5-PLANT PRODUCTS

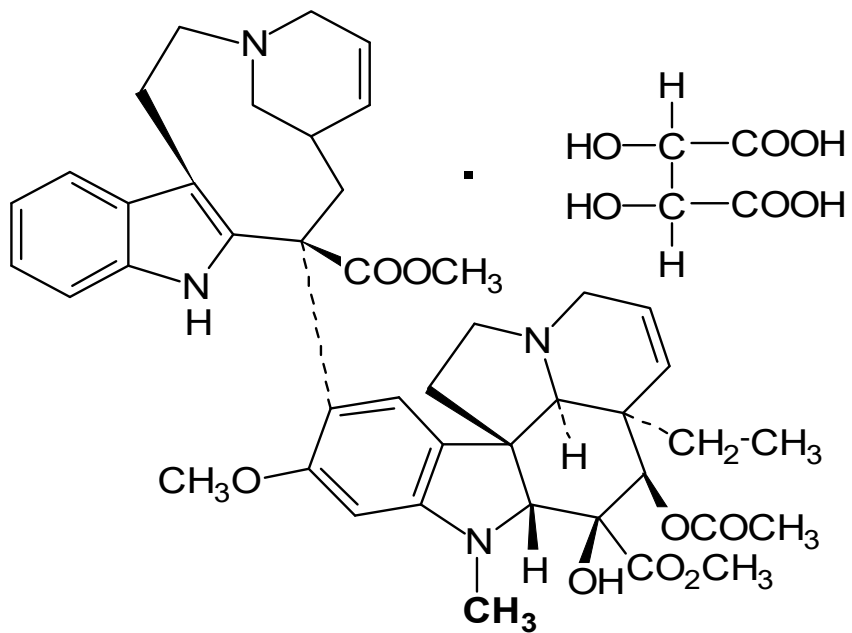
- (ANTIMITOTIC AGENTS)
- These agents prevent cellular mitosis, and in particular, interfere with the formation of the mitotic spindle.
- During mitosis; the protein tubulin undergoes **polymerization to form mitotic spindle.**
- Antimitotic agents interfere with this either by depolymerization of the microtubules or by causing structures other than the normal mitotic spindle to be formed.
- In the absence of a properly formed mitotic spindle, the chromosomes cannot correctly segregate and this ultimately leads to cellular death.

- Vinca Alkaloids: **Vincristine, Vinblastin and Vinorelbine**
 - These alkaloids are composed of two multi ringed units.
 - The vinca alkaloids contain tertiary amino groups that form salts which are freely soluble in water.



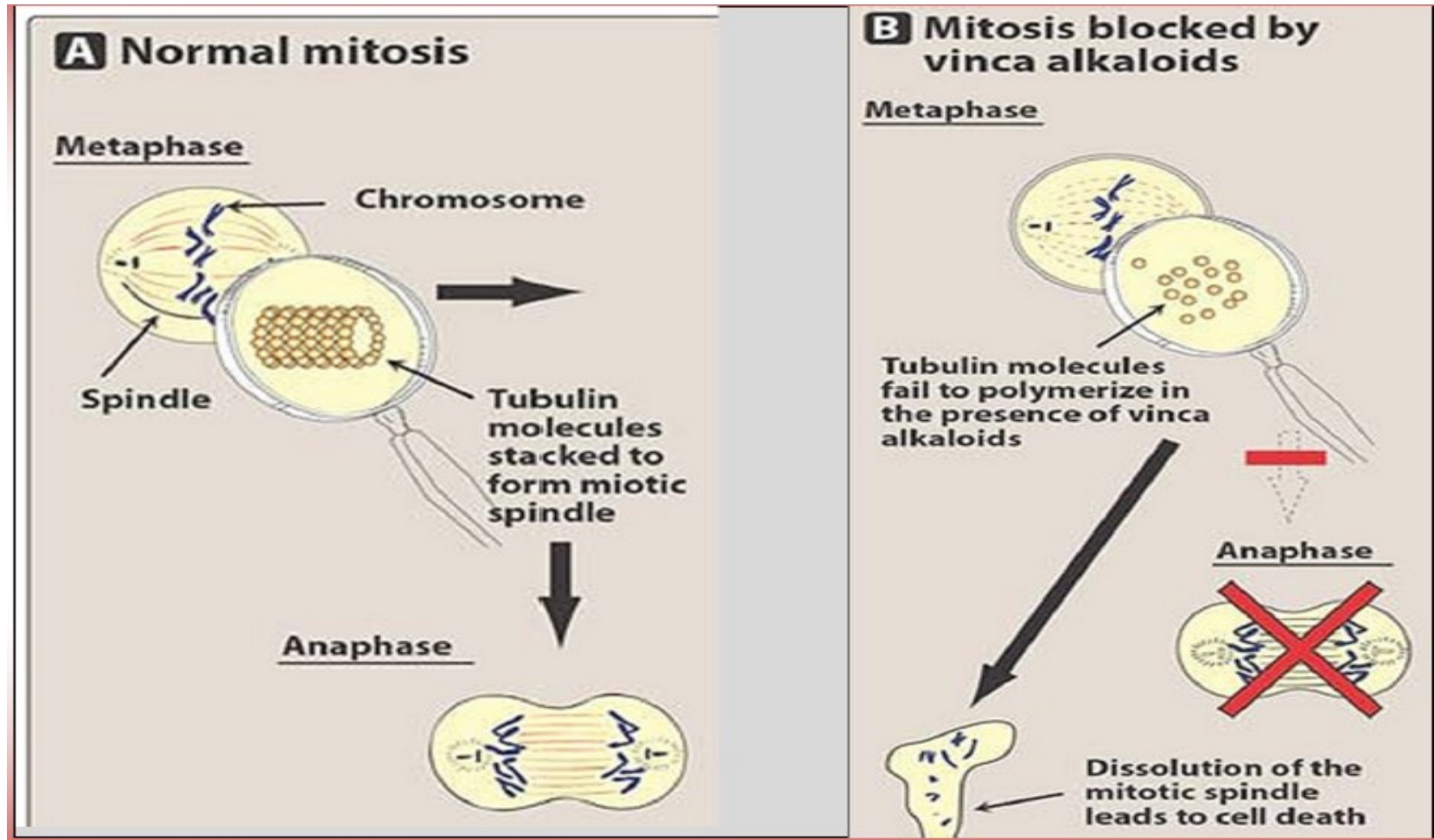
Vincristine: $\text{R} = \text{CHO}$

Vinblastin: $\text{R} = \text{CH}_3$



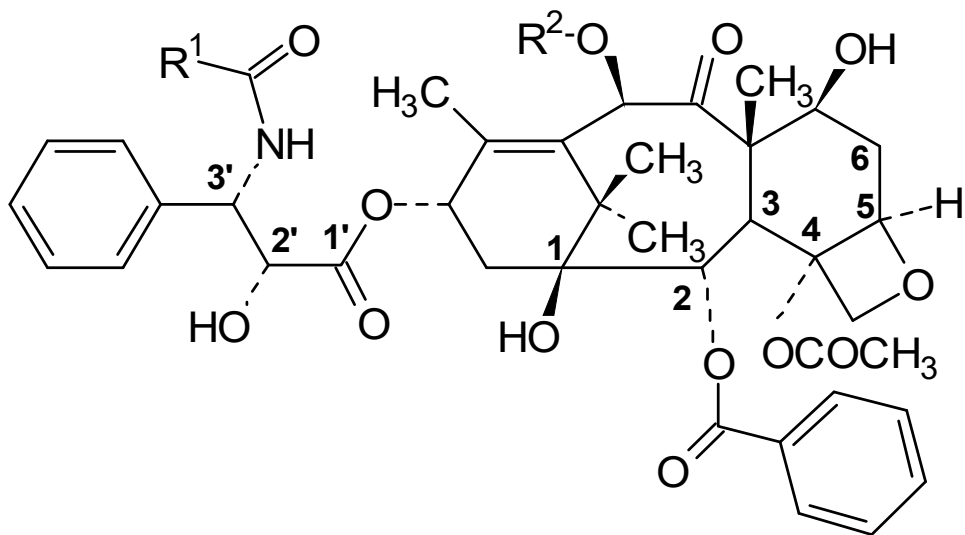
Venorelbine

Mechanism of action



ii. Taxans

- **Paclitaxel and docetaxel are taxan** derivatives of yew tree black.
- **Docetaxel** is active in patients with breast cancer resistant to anthracyclines.
- Both derivatives are excellent choices for the first and second line treatment of patients with metastatic breast cancer.
- **Paclitaxel** is administered intravenously in non aqueous solution.
- **Docetaxel** is a taxoid differing in structure from paclitaxel in having one less acetate group and a tertiary butyl carbamate group in place of the benzamido group.
- The injectable solution must be diluted prior to use.



Paclitaxel: $R^1 = C_6H_5$; $R^2 = \text{Acetyl}$

Docetaxal: $R^1 = (CH_3)_3C-O$; $R^2 = H$



6. MISCELLANEOUS ANTICANCER AGENTS

- i. Podophyllotoxins
 - They are obtained as extracts of the May apple plant, used to treat small cell carcinomas of the lung and Hodgkin's diseases.
 - They cause DNA linkages or DNA strand breaks by inhibiting topoisomerase II.
 - It may be applied topically as paint.
- ii. Hydroxyurea $\text{H}_2\text{N}-\text{CO}-\text{NH}-\text{OH}$
 - It inhibits **ribonucleotide diphosphate reductase**, thus interferes with the conversion of **ribonucleotides** to **deoxyribonucleotides** required for DNA synthesis

